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Does Carbon Regulatory Risk Reprice European Equities?

Evidence from EU ETS Market Stability Reserve Legislative Announcements

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Statement of Originality

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To my mum and dad, for their unwavering support.

Abstract

This thesis examines whether the abnormal-return response to announcements related to the EU Emissions Trading System's Market Stability Reserve (MSR) differs across firms according to their carbon intensity. Using firm-level return data for STOXX 600 constituent firms over 2013–2025, I first estimate abnormal returns from a Fama–French five-factor-plus-momentum model with rolling betas. I then partition the announcements into supply-neutral and supply-active events, separating the regulatory-risk (discount-rate) channel from the competing supply (cash-flow) channel by construction, rather than by ex-post attribution. Through a signed panel event study using 27 hand-classified announcement dates, I find no evidence that this differential departs from zero: across specifications, the 95% confidence interval bounds the effect to within roughly ± 10 basis points per standard deviation of carbon intensity, a bound no robustness specification widens by more than one basis point. With 1.9 million firm-days, the design estimates this differential precisely enough to deliver an informative null: carbon regulatory risk is not repriced at legislative announcement dates in economically meaningful magnitudes.

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1 Introduction

It would not be an overstatement to see climate change as the defining issue of the 21st Century. It is changing the way most people and society as a whole function; climate change has disrupted long-established industries and will increasingly do so. A growing body of research has begun investigating the reinforcing mechanism of financial markets both reacting and affecting this transition. Macrofinance mechanisms have been put forward to account for and price the societal cost of climate change. Led by the pioneering work of Nordhaus (1991), different models have been proposed in search for a seamless integration of the carbon cost in the market. Subsequently, climate change has also been considered through the lens of risk; financial markets' new goal being to account for this risk in their asset pricing modelling.

Physical risk and transition risk became the most used taxonomy for climate change induced risk (Giglio, Kelly, & Stroebel, 2021). Direct damage such as destruction of plant and property is easily classified as a physical risk, whereas transition risk is an umbrella term for all the risk derived from building an economy that does not have polluting substances as main drivers of growth. Given this inherent complexity, accounting for transition risk has represented a challenge both from a measuring and modelling perspective. Crucially, climate change is a stochastic process and most often investors have been deemed to underestimate the impact in their modelling. Specifically, transition risk pricing in equities has been an issue of strong contention in the past five years. Bolton and Kacperczyk (2021) ascertain that a carbon premium does exist and it cannot be explained by any other factor loadings. Subsequent research has however poked holes in the result by finding flaws in the methodology and framing the carbon premium result as sensitive to data provenance and specification (Aswani, Raghunandan, & Rajgopal, 2024; Zhang, 2025). However contentious the issue of carbon premium, institutional investors have maintained that climate risk and most urgently regulatory risk is material to their portfolio construction (Krueger, Sautner, & Starks, 2020; Stroebel & Wurgler, 2021).

One reason behind the apparent contradictory results around carbon premium might have its roots in

the design of previous research. Unconditional premium regressions cannot separate the cash-flow and the discount-rate channels which influence the stock pricing as described by Campbell and Shiller (1988). Therefore, I propose an event design mechanism that separates the two channels and analyses the regulatory news effect of the Market Stability Reserve through the theoretical lens proposed by Pástor and Veronesi (2012) and Pástor and Veronesi (2013).

The Market Stability Reserve is not defined by one event, but a decade long series of legislative announcements with various implications. It is precisely this heterogeneity that enables the separation of the two distinct channels that feed the transition risk. Specifically, some events introduced or resolved regulatory risk influencing the discount rate, while others only changed the allowance supply, hence affecting the cash flow channel. The analysis of MSR allowed for an identification design where I propose two signed variables that affect the AR. As predicted by Pástor and Veronesi (2012): risk-introducing events are therefore signed with a negative 1. Mirroring the logic of Pástor and Veronesi (2012), I also consider risk-resolving events and signed them with a positive 1, based on the intuition that when risk disappears so would the risk premium previously necessary.

Subsequently, since carbon regulation affects firms unevenly, MSR-related legislative announcements should be priced differently across firms depending on their carbon intensity. The following research question is meant to account for this heterogeneity:

> Does carbon intensity amplify the pricing effect of EU Market Stability Reserve-related legislative announcements?

Given the focus on the two distinct pricing channels, the following hypotheses follow:

1.1 Hypotheses

Hypothesis 1: Amplification The cross-sectional AR response to a signed regulatory-risk announcement increases in carbon intensity.

Hypothesis 2: Risk Channel For uncertainty introducing events, higher-carbon firms experience negative announcement AR as the market installs a risk premium (according to Pastor-Veronesi). For uncertainty-resolving events, higher-carbon firms have an increase in announcement AR as the policy risk premium collapses.

Hypothesis 3: Supply Channel On supply-active events signed by cash-flow direction (supply sign), the carbon-scaled announcement response is non-zero. Where the direct-cost effect dominates, a supply tightening lowers high-carbon firms' returns and a loosening raises them; the placebo's magnitude bounds the cash-flow channel's contribution against the clean-risk result.

To test the aforementioned hypotheses, I conducted a panel event study regression on the STOXX 600 historical membership between 2013-2025 with 27 events being hand identified as a milestone in the legislative process by either influencing the supply channel, the discount factor channel or both. The design uses date fixed effects such that any market-wide response is absorbed and therefore this research only looks at any residual risk that is being priced differently based on carbon intensity during MSR-related legislative announcements.

First, abnormal returns are derived using the Fama-French five-factor plus momentum with rolling beta. After deriving the abnormal returns the variable of interest is the +/-1 event sign interacting with the within-year carbon intensity z-scores. Crucially, the channel-separation architecture is the following: supply-neutral events identify the risk channel, while re-signing supply-active events by cash-flow direction tests the competing channel. Identification comes from cross-sectional differentiation within event dates, which rules out market confounds by construction.

I find no differential carbon response across any specification, with the 95 percent confidence interval ruling out carbon-scaled effects for both the main discount risk channel and supply channel beyond roughly ± 10 bp per standard deviation in the announcement window on the two-way-clustered specifications. The results remain robust even with changing the rolling beta window from 48 months to 60, controlling for energy prices or dividing the cross-sectional in only a binary high and low carbon intensity specification. While the clean-risk pre-event coefficient is marginally

significant under two-way clustering; decomposition attributes it entirely to the resolving events, where the estimate's apparent precision is a firm-clustering artefact with the significance fading under different specifications.

An announcement window measures the repricing that occurs when residual outcome uncertainty resolves, and nothing that has already been incorporated. That distinction is central here, because the MSR is a forecastable rule whose milestones are calendar-scheduled and whose mechanics are deterministic. Scheduling does not place such events outside the framework: Kelly, Pástor, and Veronesi (2016) show that options whose lives span known political event dates are systematically more expensive, evidence that outcome uncertainty is priced at a scheduled date even when the timing carries no surprise. What the design identifies is therefore the residual, a different object from the continuously priced premium estimated in the unconditional cross-section (Bolton & Kacperczyk, 2021; Hsu, Li, & Tsou, 2023; Oestreich & Tsiakas, 2015), and four forces compress it. Two operate on the estimand and are economic: partial anticipation, since a signed event indicator fires at full weight however much of the news was already priced, and the regime's finite expected horizon before the scheduled ETS review, which scales down the revision any single announcement can deliver. Two are econometric: measurement error in vendor-supplied emissions, and misclassification of the resolving events, whose $+1$ coding is the design's most fragile assumption. All four push the estimate toward zero, which is why I scope the bound to residual announcement-window repricing rather than to the structural premium, and why the sign-unambiguous introducing events are reported separately: that subset removes the sign attenuator by construction. The bound is consistent with evidence that realised carbon-related return differentials reflect gradual revisions in climate and policy concern rather than discrete legislative milestones (Pástor, Stambaugh, & Taylor, 2022).

Notably, as better carbon intensity data reliability will become more common in future years, results will have increasingly narrower margins of error and will pinpoint the effect if any much better than the current state of the data.

This thesis makes three contributions. First, to identification: the regulatory-risk and cash-flow channels of EU ETS news are separated *ex ante*, in the construction of the treatment itself with supply-neutral events identifying the discount-rate channel, while supply-active events re-signed by their cash-flow direction supply the competing mechanism's own test, where the existing literature separates the channels ex post, by return decomposition (Ardia, Bluteau, Boudt, & Inghelbrecht, 2023) or by splitting the reacting sample (Hengge, Panizza, & Varghese, 2023). Second, to measurement: across roughly 850 firms and nearly two million firm-day observations, the resulting null is an informative bound, confining the residual carbon-scaled announcement repricing within roughly ± 10 basis points per standard deviation of carbon intensity in every specification for which standard errors are reliable. Third, to theory: the object this bound disciplines is the cross-sectional amplification implied by the Pástor–Veronesi framework (Pástor & Veronesi, 2012) as operationalised by the applied climate-finance literature (Hengge et al., 2023; Ramelli, Wagner, Zeckhauser, & Ziegler, 2021), repricing that loads on carbon intensity at announcement, and it is that operationalisation which fails to manifest at announcement frequency in this setting.

The remainder of this thesis is organized as follows. Section 2 presents the institutional background of the EU ETS and the Market Stability Reserve and introduces the taxonomy of legislative events on which the identification rests. It also provides a breakdown and critique of the economic theory and the desired effects of the introduction of the MSR. Section 3 reviews the literature on political uncertainty and asset prices, the carbon premium and its channels, and climate policy event studies, and derives the three hypotheses; Section 4 develops the theoretical mechanism linking regulatory-risk announcements to abnormal returns; Section 5 describes the data; Section 6 specifies the empirical methodology, Section 7 reports the main results and robustness checks, Section 8 details policy implications and direction for further research, and Section 9 concludes. Notably, the use of AI throughout the project is documented in Appendix A.

2 Institutional Background

2.1 The EU Emissions Trading System: History and Phases

The EU ETS is the European Union's principal carbon-pricing instrument and the world's first large-scale cap-and-trade system for greenhouse gases: covered firms must surrender one allowance for every tonne of CO_2-equivalent they emit, against a capped total supply. Its form was itself a product of political constraint: an EU-wide harmonised carbon tax proved politically infeasible, since taxation remained a sovereign competence of member states, and a trading system offered the same aggregate quantity discipline while leaving allocation and revenues at the national level (Borghesi, Pahle, Perino, Quemin, & Willner, 2023; Ellerman & Buchner, 2007). Established by Directive 2003/87/EC, the system launched in 2005, covering electricity generation and energy-intensive industry, roughly half of the EU's greenhouse-gas emissions (Borghesi et al., 2023), and has since operated in four trading phases, each defined by its cap-setting architecture and each shaped by the failures of the one before.

Phase 1 (2005–2007) was an explicit pilot. Caps were set bottom-up through National Allocation Plans, allowances were allocated free of charge, and banking into the next phase was prohibited. The design failed quickly: when verified emissions data released in April 2006 revealed that allocations exceeded emissions, the allowance price halved within days and, with no banking to carry value forward, fell to zero by 2007 (Bushnell, Chong, & Mansur, 2013; Ellerman & Buchner, 2007). That April 2006 collapse is also the methodological origin of this thesis's event-study lineage, as Section 3.3 develops.

Phase 2 (2008–2012) coincided with the Kyoto commitment period. Caps tightened, banking into subsequent phases was permitted, and international credits could be surrendered for compliance. The binding problem was no longer design but demand: the 2008 financial crisis, overlapping renewable-energy policies, and the credit inflow generated a structural surplus of allowances that the fixed cap could not absorb, and prices persisted in the range of EUR 4–10 from late 2011 (Borghesi

et al., 2023). By 2018 the accumulated surplus amounted to approximately 140% of annual ETS emissions (Rosendahl, 2019). At these prices it could no longer be argued that the mechanism was doing what it was built to do: give firms an incentive to cut their emissions.

Phase 3 (2013–2020) replaced the bottom-up architecture with a single EU-wide cap declining at a linear reduction factor of 1.74% per year, made auctioning the default allocation mode for the power sector, and harmonised residual free allocation through product benchmarks. The surplus, however, was inherited, and the first response to it failed informatively: the backloading of 900 million allowances from the 2014–2016 auctions, to be returned in 2019–2020, had no sustained price effect, exactly as intertemporal arbitrage predicts for a measure that shifts the timing of supply without changing its total (Koch, Fuss, Grosjean, & Edenhofer, 2014; Salant, 2016). That failure motivated a structural remedy: the Commission proposed the Market Stability Reserve in January 2014, which is the first event in this thesis's registry, and Decision (EU) 2015/1814 established it. The original MSR was itself cap-neutral, withholding allowances only to release them later, and was accordingly predicted, and observed, to have limited price impact (Perino & Willner, 2016); its legality was moreover contested by Poland before the European Court of Justice, a challenge whose filing and resolution supply two of the clean-risk events studied below.

The decisive change came with Directive (EU) 2018/410, which doubled the reserve's intake rate and, from 2023, permanently invalidates allowances held in the reserve above a threshold. This made the long-run cap endogenous to market outcomes and, in Perino's (2018) phrase, punctured the waterbed: under a fixed cap, any demand-reducing shock merely displaces allowance use in time, whereas under the invalidation rule it can reduce cumulative supply (Perino, 2018; Rosendahl, 2019). Regulatory events thereafter carry genuine supply content which is the property on which this thesis's supply-active event classification rests. Phase 4 (2021–2030) then raised the linear reduction factor to 2.2%, and the Fit for 55 revision enacted by Directive (EU) 2023/959 tightened the trajectory further with a 62% emissions reduction in ETS sectors by 2030, a reduction factor of 4.3% rising to 4.4%, and a fixed 400-million invalidation threshold, while extending the strengthened MSR parameters to 2030.

The estimation window of this thesis, 2013–2025, therefore spans Phase 3 in its entirety and Phase 4 through 2025: the period in which the MSR was proposed, legislated, contested, reformed, and operated, and in which auctioning, not free allocation, was the power sector's allocation mode throughout. The 2010–2012 years, which lie in Phase 2 and are contaminated for analysis by the backloading episode, serve only to warm up the rolling beta estimator and never enter the results (Section 6.1). All twenty-seven events in the registry, 2014–2025, fall inside the two phases studied.

2.2 The Market Stability Reserve

The MSR was introduced to address the imbalance between the supply of and demand for allowances that Phase 2 bequeathed. By adjusting the supply of allowances, it moves expectations about future carbon prices, and with them the expected cost of compliance for carbon-intensive firms, the channel through which reserve-related news can reach equity values at all.

Established by Decision (EU) 2015/1814 and operational since January 2019, the MSR is a rules-based mechanism that adjusts the volume of allowances auctioned in response to a single quantity indicator: the total number of allowances in circulation (TNAC), the stock of allowances banked by market participants (cumulative supply less cumulative surrender and the reserve's own holdings) which the Commission publishes each year (Borghesi et al., 2023). The reserve operates in one of three modes. If the published TNAC exceeds an upper threshold of 833 million allowances, a fraction of the TNAC, the intake rate, is withheld from the following year's auctions and placed in the reserve; if the TNAC falls below 400 million, 100 million allowances are released from the reserve back to auctions over twelve months; between the thresholds the reserve is idle (Decision (EU) 2015/1814). The intake rate was set at 12% and doubled to 24% by Directive (EU) 2018/410, an elevated rate since extended through 2030 together with a 200-million minimum intake (Directive (EU) 2023/959). Because each year's withholding is a deterministic function of a published number, the MSR's operation is, by construction, forecastable: the annual TNAC publication resolves the reserve's next action, which is why those publications enter the event registry as recurring,

informationally thin dates rather than as regulatory news.

Two subsequent amendments changed what the reserve *is*. First, Directive (EU) 2018/410 added the invalidation provision: from 2023, allowances held in the reserve above a threshold are permanently cancelled rather than stored. This is the puncture of the waterbed described in Section 2.1 it converts the reserve from a device that re-times supply into one that destroys it, making the long-run cap endogenous to market outcomes (Perino, 2018). The first such invalidation, on 1 January 2023, cancelled approximately 2.5 billion allowances, the largest single supply removal in the system's history; the annual invalidations that followed are the supply-placebo events of this thesis. Second, Directive (EU) 2023/959 recalibrated the mechanism within the Fit for 55 revision: the invalidation threshold was fixed at 400 million allowances from 2024 (replacing the earlier auction-volume reference), a 100% marginal intake rate now applies when the TNAC lies between 833 and 1,096 million to remove oscillatory behaviour around the upper threshold, and aviation allowances entered the TNAC calculation (Borghesi et al., 2023).

Two properties of this design matter for identification. The mechanism's *operation* is mechanical, but its *rules* were legislated, contested, and revised over a decade: proposed in 2014, adopted in 2015, challenged before the European Court of Justice, reformed in 2018 and again in 2023. News about the rules changes the regulatory environment without moving the allowance supply on the announcement date; the mechanism's scheduled operations move supply without conveying news about the rules. It is exactly this separation, between announcements about what the reserve *will be* and mechanical applications of what it *already is*, that the event taxonomy of the next subsection formalises.

2.3 Key Events

Because the MSR is not a single-moment intervention but a decade-long legislative process, this thesis identifies twenty-seven impactful events over 2014–2025, hand-classified from the legislative record. Each event is characterised along the two dimensions the previous subsection distin-

guishes: whether it moves the *regulatory-risk* environment, introducing uncertainty about the future carbon-pricing regime or resolving it, and whether it moves the expected *allowance supply*. The Commission's first proposal of the MSR on 22 January 2014 introduces regulatory risk, placing an untested regime on the table, while also signalling tighter future supply; the formal adoption of Decision (EU) 2015/1814 resolves the open legislative question, though it too carries supply content. Events that load on only one dimension are the analytically valuable ones: a legal challenge to the reserve moves regulatory risk with no supply content, while the annual invalidations execute supply changes under rules already known, carrying no regulatory news. The TNAC annual publications, finally, are recurring and informationally thin, the mechanical resolution of a deterministic rule, and, together with implementation milestones such as the reserve becoming operational in 2019, receive no event window. The role of this classification is to distinguish the types of informational shock, the direction in which each should move abnormal returns, and the channel through which it does so; its formalisation into two signed variables and analysis groups, with the full event listing, is developed in the data section (Section 5, Table 2).

3 Literature Review

3.1 Political Uncertainty and Asset Prices

That policy uncertainty must affect asset prices is close to an axiom, measuring and understanding the mechanism is another matter entirely. Pástor and Veronesi (2012) develop a model in which investors learn about the impact of the prevailing policy from firms' realized profits, and the government is more likely to replace a policy that appears harmful. In this setting policy changes command a risk premium, larger in weaker economic conditions, and the average stock-price reaction to the adoption of a change is negative. (Crucially, this repricing operates through the discount rate: policy uncertainty moves prices even absent any revision to expected cash flows.) It follows that a direct cross-sectional corollary is that firms more exposed to the policy under revision

bear more of that priced risk and reprice more sharply when it resolves.

A natural objection is that the PV mechanism requires unexpected events, whereas regulatory votes and Council decisions sit on a known legislative calendar. Kelly et al. (2016) answer this directly: studying options around scheduled elections and summits, they show the framework applies to anticipated events because what is priced is the resolution of residual outcome uncertainty, not the timing of the event itself: options protecting against price moves are more expensive around such votes, and more so when the outcome is less certain.

These pricing predictions have direct empirical support. Brogaard and Detzel (2015) show that economic-policy-uncertainty innovations carry a significant risk premium — the stocks most exposed to policy uncertainty underperform the least exposed by over five percent annually — and, crucially for the mechanism at issue, that this operates through the discount rate rather than through cash flows; the aggregate policy-uncertainty index they employ is that of Baker, Bloom, and Davis (2016).

3.2 The Carbon Premium and its Channels

Bolton and Kacperczyk (2021) found higher average return for high-emission firms, empirical evidence that there is a carbon premium or, equivalently, what it is commonly known in climate finance as 'greenium': green firms have lower expected returns. Furthermore, they find that carbon premium cannot be explained through a sin divestment effect. The premium sits on an emission level and growth, while divestment operates on a carbon intensity level, a first indication, developed in Section 3.4 that the result depends on the carbon proxy.

The failure of observable screening to explain the premium does not, however, rule-out demand based explanations. Investors not only account for profitability, but also derive utility from investing in a socially responsible manner (Hong & Kacperczyk, 2009; Riedl & Smeets, 2017). What Riedl and Smeets (2017) further note is that the reasoning for social investing is both intrinsic and extrinsic; some investors only want to hold ESG assets for signalling purposes and if the reasoning behind the

signalling disappears, the investors' taste would also fluctuate affecting the amount of divestment. The existence of carbon premium is therefore contingent not only to how investors account for climate risk, but equally to how investors' taste changes. Pástor, Stambaugh, and Taylor (2021) show that green taste lower green firms' expected return and raise brown firms' in equilibrium, but, crucially, green assets outperform when positive shocks affect the ESG factor, an indication that as taste changes, the premium becomes a moving target.

Pástor et al. (2022) finds that the realised green outperformance over the 2010's reflected the unexpected strengthening of climate-conscious investors rather than any confirmation of the equilibrium premium. Ardia et al. (2023) also confirms Pástor et al. (2021) prediction on a 2010-2018 sample. Using a daily media-based climate concern index, they find that on days with unexpected increase of climate concerns, prices for green stocks increase. This shift in demand is taste-based, suggestive of a discount rate mechanism reflecting investors' need for a premium in order to hold "sin stocks". As such the realised premium becomes a function of the concern process which is politically and attention-contingent. Moreover, measured green demand partly follows labels rather than fundamentals: The United Nation Principles for Responsible Investments signing attracts flows without corresponding portfolio changes among U.S. institutions (Gibson Brandon, Glossner, Krueger, Matos, & Steffen, 2022; Kim & Yoon, 2023). The regulatory response such as SFDR fund classification and ESMA's fund-naming rules therefore reshuffles which assets the green demand attaches to, so premium estimates spanning these regimes partly reflect relabeling rather than repricing.

With multiple determinants of premium instability, analysing premium-level without any separation of channels only provides the snapshot of a moment rather than definitive proof of a premium. Understanding if the premium is a product of firms becoming riskier via a discount rate channel or because regulation damages their cashflow calls for more research aimed at channel separation. While, Ardia et al. (2023) do account for channel separation, their analysis does the separation by statistical attribution rather than by construction. With MSR, the channel separation is inherent in the way legislative announcements for MSR are made that provides clean only supply channel and

only risk channel events.

3.3 Climate Policy Events and EU ETS

Political and legislative events offer a direct route to measuring the pricing of climate regulatory risk, but political events rarely move one policy dimension at a time, so event selection is crucial. Ramelli et al. (2021) provide a conceptual template: the 2016 and 2020 U.S. elections serve as opposite-signed shocks to expected climate-policy stringency, crossed with firms' climate characteristics in each event's cross-section. The opposite-shock design disciplines interpretation, since a pre-existing trend toward responsible investing cannot flip sign with the shock. Yet the treatment remains regime-wide, moving expected cash flows and regulatory risk simultaneously; the authors separate the channels only ex post, inferring a discount-rate effect from what analysts' earnings-forecast revisions fail to explain. This thesis instead builds the separation into the research design: pooling many legislative events under an explicit sign variable and partitioning them ex ante into supply-neutral events, which move regulatory risk alone, and supply-active events, which also shift expected cash flows.

The cash-flow mechanics of the ETS are laid bare by Bushnell et al. (2013). When leaked, compliance data revealed the Phase 1 over-allocation in April 2006 and the allowance price halved, equity values fell across carbon and electricity-intensive industries despite the reduction in compliance costs, implying that investors price ETS supply news through expected profits and product prices, not compliance costs alone. Remarkably, they show that within the power sector the effect is exactly the reverse of what would be expected: the firms with the lowest carbon intensity saw their stock prices decline the most. The explanation is that the carbon price passes through the marginal fossil plant's bid into the wholesale power price, pass-through that electricity's highly inelastic demand makes nearly complete, so low-carbon generators collect it as pure margin. That rent evaporated with the carbon price crash, while dirtier generators' losses were partly offset by cost relief. The free allocation to power generators that produced the windfall they document ended with full auctioning

at the start of Phase 3 in 2013; the pass-through mechanism, however, is allocation-independent, so the power-price channel and the clean generators' inframarginal rent persist throughout the sample period studied here. The cash-flow channel is thus demonstrably powerful, and its loading on carbon intensity is not even sign-stable across sectors: inverted within power, negative for EU-exposed industry. Hence, any event carrying supply content contaminates a carbon-scaled coefficient in a direction that cannot be signed a priori, a strong rationale for the supply-neutral partition.

Känzig (2023) establishes ETS regulatory news as a source of identified policy shocks. From high-frequency EUA-futures reactions around 114 regulatory events concerning the supply of allowances, he constructs a carbon policy surprise series and traces its macroeconomic effects: energy prices rise, emissions fall persistently, activity contracts, and aggregate stock prices decline by over 2 percent. His identifying variation is thus, by construction, allowance-supply news. His design harnesses the cash-flow channel for identification, whereas this thesis' design removes it to isolate regulatory risk. Beyond the channel contrast, the designs differ econometrically in two respects: the outcomes are aggregates rather than the firm cross-section, and the event-count asymmetry follows from the selection criteria. As such, his design harvests events in bulk because the market signs them. By contrast, the channel partition in this thesis pays for channel identification with scarcity.

The nearest design to this thesis is Hengge et al. (2023), who adopt and extend the event registry of Känzig (2023) with 98 supply-related regulatory events over 2011–2021 worked into the firm-level cross-section: daily EUA-futures surprises on event days interacted with lagged emission intensity for over 2,000 listed European firms. They find that surprises raising the carbon price produce negative returns increasing in carbon intensity, and they pose the channel question directly, distinguishing a direct input-cost channel from a transition-risk channel. Their separation strategy is a sample split on the outcome side: effects strengthen when ETS-covered sectors are dropped, which they read as investors pricing transition risk. Yet the treatment itself remains supply news which bundels cost and resolve components for every firm. Furthermore, non-ETS firms are not free of the cost channel, since carbon passes through electricity prices to Scope 2-intensive firms (see Section 2.3's discussion of Bushnell et al., 2013), whereas the design of this thesis removes supply

content from the treatment itself: separation by what the news contains rather than by who reacts.

3.4 Measurement and Fragility

As I've hinted at in section 3.2, the existence of carbon premium is still an issue of open debate in the literature. In their response to Bolton and Kacperczyk (2021) carbon premium empirical finding, Aswani et al. (2024) lay bare how fragile the carbon premium is. Firstly, the premium appears in vendor-estimated emissions and disappears when the sample is restricted to firms that disclose, and because vendor estimates are largely imputed from firm financials, an association with estimated emissions partly reflects fundamentals rather than priced carbon exposure. Secondly, the premium is found in emission levels rather than intensity; since total emissions scale with firm size, and size and carbon-intensity both vary by sector, the levels premium partly conflates carbon content with an industry effect.

Return construction is equally decisive. Zhang (2025) shows that emissions are released to investors with long publication lags yet are mechanically tied to contemporaneous sales, so relating returns to insufficiently lagged emissions imports future-performance information: once emissions are lagged to their true availability and firm sales are controlled for, the premium turns negative in the United States and insignificant everywhere else.

Beyond data and construction, the premium is fragile to when it is measured: as I argued in Section 3.2, the taste process generating it is non-stationary (Pástor et al., 2022). Therefore, any sample-period estimate partly reflects the realized path of demand shocks rather than a stable equilibrium object.

Furthermore, even a premium robust to all of the above measurement still does not reveal why it exists: level estimates conflate the regulatory-risk and cash-flow channels, and the separation attempts in the literature (return decomposition, forecast proxies, outcome-sample splits) each rest on a modelling or classification assumption rather than on treatment that isolates a channel by construction.

Against this fourfold fragility, I argue that the informative object is not another unconditional premium estimate but a precisely bounded response to identified regulatory news in a design whose every assumption, be it event selection, sign coding or emission measurement, is open to inspection.

3.5 Hypothesis Development

The research question of whether carbon intensity amplifies the pricing of MSR-related legislative announcements translates into three hypotheses. Each is a prediction about the cross-sectional relationship between a firm's carbon intensity and its abnormal return around an event. Because events are signed by the direction of the shock, the predictions below are stated first in terms of raw abnormal returns for each event sub-type, and then expressed as the sign of the coefficient β on the interaction $\text{sign}_t \times \text{carbon}_{z,it}$ estimated by the signed specification in Section 6, where $\text{carbon}_{z,it}$ is the firm's within-year standardised carbon intensity and $\text{sign}_t \in \{-1, +1\}$ signs the event by the direction of the regulatory-risk shock. The coefficient is scaled so that $\beta > 0$ places high-carbon firms in the Pástor–Veronesi-predicted direction: repricing that amplifies with policy exposure.

From Section 3.1 (political uncertainty and asset prices), Pástor–Veronesi framework prices regulatory uncertainty through a discount-rate channel, and its cross-sectional corollary is that firms more exposed to the policy under revision reprice more sharply when that uncertainty resolves. Taking carbon intensity as the measure of exposure to EU ETS regulatory risk yields the baseline hypothesis.

> **Hypothesis 1: Amplification.** The cross-sectional abnormal-return response to a signed regulatory-risk announcement increases in carbon intensity: $\beta > 0$ on the pooled signed events.

From Sections 3.1 and 3.2 it becomes clear that Hypothesis 1 does not identify why high-carbon firms reprice: both discount-rate repricing and cash-flow revision imply $\beta > 0$ once events are signed, since a tightening that raises regulatory risk also raises expected compliance costs. Section

3.2 established that separating the two requires news that moves one channel while holding the other fixed. The supply-neutral (clean-risk) events revise the perceived path of regulatory risk without altering the contemporaneous supply of allowances, so any carbon-scaled repricing they induce is attributable to the discount-rate channel.

> **Hypothesis 2: Risk channel.** On supply-neutral events: at uncertainty-introducing events higher-carbon firms earn more negative announcement returns as a risk premium is installed; at an uncertainty-resolving event, higher-carbon firms earn more positive returns as that premium collapses. Because introducing events are coded -1 and resolving events $+1$ (`pv_ar_sign`), both sub-predictions correspond to a single positive coefficient, $\beta > 0$, in the clean-risk specification (Table 4). This is the primary identifying test.

In Sections 3.2 and 3.3, I described how the competing channel operates through expected profits: allowance-supply news transmits to equity values through product prices and compliance costs. To test whether this channel is independently priced in the cross-section, the supply-active events are re-signed by their cash-flow direction (`supply_ar_sign`) rather than their risk direction, the competing mechanism's own treatment. Section 3.2 showed, however, that the cash-flow channel's loading on carbon intensity is not sign-stable across sectors: cost pass-through inverts it within the power sector. The directional prediction therefore holds only where the direct compliance-cost effect dominates, and the hypothesis is stated as a two-sided test.

> **Hypothesis 3: Supply channel.** On supply-active events signed by cash-flow direction (`supply_ar_sign`), the carbon-scaled announcement response is non-zero: $\beta \neq 0$ in the supply-placebo specification (Table 4). Where the direct-cost effect dominates, a supply tightening lowers high-carbon firms' returns and a loosening raises them; the placebo's magnitude bounds the cash-flow channel's contribution against the clean-risk result.

The classification and signing of MSR events into supply-neutral and supply-active sets, the taxonomy on which the H2/H3 partition rests, is previously developed in the institutional-background section and rationalised in the data section.

4 Theoretical Framework

This section develops the theoretical foundation for the empirical design. The regulatory-risk channel is anchored in the general-equilibrium model of Pástor and Veronesi (2012), in which government policy decisions move stock prices through the stochastic discount factor rather than through expected cash flows. The framework delivers three results that the empirical design operationalises: policy announcements carry a discount-rate effect distinct from any cash-flow effect; the discount-rate effect is priced in the cross-section in proportion to firms' exposure to policy impact; and the magnitude of the announcement response depends on how much outcome uncertainty remains unresolved at the announcement. The competing cash-flow channel, through which allowance-supply decisions move the expected profits of carbon-intensive firms directly, is anchored in Bushnell et al. (2013) and is deliberately excluded from the primary event set by construction, as Section 6 details.

4.1 Policy uncertainty and firm profitability

In the Pástor and Veronesi (2012) economy, each firm i's profitability follows

$$d\Pi_t^i = (\mu + g_t)\, dt + \sigma\, dZ_t + \sigma_1\, dZ_t^i, \tag{1}$$

where g_t is the impact of the prevailing government policy on mean profitability, Z_t is an aggregate shock, and Z_t^i is idiosyncratic (their equation (1)). The policy impact g_t is constant while a given policy is in force and is unknown to all agents, including the government. Prior beliefs about the

impact of any policy, be it incumbent or replacement, are $g \sim N(0, \sigma_g^2)$, where σ_g is labelled *impact uncertainty* (their equation (3)). A second, distinct source of randomness is *political uncertainty* σ_c: the government's policy decision depends on a political cost c that investors never observe, so even an investor who knows everything else cannot fully anticipate the decision. The distinction matters for the mapping to the EU MSR below: impact uncertainty concerns what a policy will do once in force, while political uncertainty concerns whether and in what form the decision will be taken.

Because the incumbent policy operates continuously through equation (1), realised profitability is a flow of noisy signals about its impact, and agents learn in a Bayesian fashion. The posterior at time t is

$$g_t \sim N\big(\hat{g}_t,\, \hat{\sigma}_t^2\big)\,, \qquad \hat{\sigma}_t^2 \;=\; \frac{1}{\dfrac{1}{\sigma_g^2} + \dfrac{t}{\sigma^2}}, \tag{2}$$

combining their equations (8) and (10): $\hat{g}_t$ is the posterior mean which is the market's running estimate of the incumbent policy's impact and $\hat{\sigma}_t^2$ is the posterior variance, which declines deterministically as data accumulate. Learning is therefore an accumulating stock of precision about the policy in force. This stock is the model's central asset-pricing state: a policy change discards it, replacing the sharpened posterior $N(\hat{g}_\tau, \hat{\sigma}_\tau^2)$ with the diffuse prior $N(0, \sigma_g^2)$, since nothing about a not-yet-operating policy can have been learned from realised profitability.

4.2 The pricing kernel and the discount-rate effect

Under market completeness the state price density is uniquely

$$\pi_t \;=\; \frac{1}{\lambda}\, E_t\big[B_T^{-\gamma}\big]\,, \tag{3}$$

their equation (16), where B_T is aggregate terminal capital and $\gamma > 1$ is relative risk aversion, so that $B_T^{-\gamma}$ is the marginal utility of terminal wealth. Any stock is priced against this kernel through the formula $M_t^i = E_t[(\pi_T/\pi_t)\, B_T^i]$ (their equation (17)); the discounting object is the ratio π_T/π_t,

which represents the stochastic discount factor property ,of which π_t is the level. Written this way, the model makes the two transmission channels of any policy event explicit and mutually exclusive in origin: an event can move prices by changing the expected payoff B_T^i (a cash-flow effect) or by changing the kernel against which that payoff is discounted (a discount-rate effect).

A policy change moves the kernel. Replacing a learned-about policy with an untested one resets posterior uncertainty from $\hat{\sigma}_\tau$ up to σ_g, which raises the volatility of the stochastic discount factor: Pástor and Veronesi (2012) show in their Proposition 6 that π_t jumps at the announcement and that its diffusion volatility is increasing in posterior uncertainty, so prices fall as investors demand compensation for the less certain environment. The cash-flow effect typically works in the opposite direction, since policies are replaced when their perceived impact is unfavourable, so the replacement raises expected profitability. Pástor and Veronesi (2012) prove (their Propositions 4 and 5) that the discount-rate effect dominates unless the outgoing policy is perceived as sufficiently harmful: the expected announcement return conditional on a policy change is negative. The empirically relevant statement of the mechanism for this thesis is therefore not merely that policy news moves prices, but that a well-identified regulatory event moves prices through π_t even when its direct cash-flow content is zero. This is precisely the property the supply-neutral MSR event set is constructed to isolate.

These level results establish the mechanism but are not the estimand. In the empirical specification below, any component of the announcement return common to all firms on an event date is absorbed by date fixed effects, so the propositions of this subsection motivate the design without being tested by it. What the design tests is the cross-sectional footprint of the mechanism, to which the next subsection turns.

4.3 Heterogeneous policy exposure and the cross-sectional prediction

The results above concern the representative firm and thus the *level* of the announcement return. The empirical design in this thesis, by contrast, is cross-sectional: it asks whether high-carbon firms

respond differentially. The relevant extension is Section V of Pástor and Veronesi (2012), which replaces equation (1) with

$$d\Pi_t^i = (\mu + \beta^n g_t)\, dt + \sigma\, dZ_t^n + \sigma_1\, dZ_t^i, \tag{4}$$

their equation (55), where firms in sector n carry a "government beta" β^n scaling their loading on the policy impact. Firms with larger β^n inherit proportionally larger exposure to revisions in $\hat{g}_t$ and to the uncertainty resets that drive the kernel, and Pástor and Veronesi (2012) show that larger differences in government betas generally translate into larger differences in expected returns, with the premium for policy exposure positive on average though state-dependent.

One feature of these results must be stated explicitly, because it separates what the model supplies from what this thesis infers. The Section V quantities Pástor and Veronesi (2012) report are differences in conditional *expected returns* between high- and low-exposure sectors, $\mu_t^H - \mu_t^L$: risk premia accruing over time, that is, differences in the level of expected returns between exposure classes, not the cross-section of returns at the announcement itself. The announcement-window prediction tested in this thesis is therefore an implication drawn from the loading structure of equation (4) rather than a result the paper states: a firm loading β^n on g_t inherits β^n times the exposure both to revisions in beliefs at the announcement and to the uncertainty reset of their Proposition 6, so the repricing the model delivers for the representative firm scales across firms with their policy exposure. The framework is used here in exactly that capacity: the model implies, rather than shows, the cross-sectional announcement response.

The identifying assumption connecting this structure to the data is that carbon intensity proxies for the policy beta with respect to EU carbon regulation: a firm whose production embeds more Scope-1 emissions per unit of revenue loads more heavily on the profitability impact of the carbon-pricing regime, whatever that impact turns out to be. Under this assumption the model's cross-sectional prediction is directional and signed. An event that raises regulatory uncertainty (or shifts the expected stringency of the regime adversely) depresses high-carbon firms relative to low-carbon

firms; an event that resolves uncertainty relieves them relative to low-carbon firms. The empirical counterpart is the interaction coefficient in

$$AR_{it} \;=\; \beta\left(\text{sign}_t \times \text{carbon}_{z,it}\right) \;+\; \gamma\,\text{carbon}_{z,it} \;+\; \alpha_i \;+\; \delta_t \;+\; \varepsilon_{it}, \tag{5}$$

where $\text{sign}_t \in \{-1, +1\}$ encodes the direction of the regulatory-risk shock and a positive β places high-carbon firms in the amplification direction the model predicts. Hypothesis H2 is the statement that $\beta > 0$ on the supply-neutral event set; the construction of sign_t is taken up next.

4.4 Mapping the model to MSR events: introducing and resolving shocks

The mapping from the model's single announcement date τ to the MSR legislative record requires care in three respects.

Firstly, the model's announcement is the moment a policy *change* occurs and uncertainty jumps up. MSR events are not all of this type. Events classified as *introducing* regulatory risk, such as a proposal or escalation that places a new or more uncertain regime on the table, correspond to the model's uncertainty-raising direction: the market's effective uncertainty over the future carbon-pricing environment rises, the kernel channel operates as in their Proposition 6, and high-carbon firms should underperform, motivating the sign coding -1. Events classified as *resolving*, such as a final vote or adoption that collapses a previously open question, are the mirror image: outcome uncertainty falls rather than rises, the discount-rate effect operates in reverse, and high-carbon firms should be relieved, motivating $+1$. The signed design thus does not force every event into the mould of the model's τ; it uses the model's mechanism symmetrically, with the sign carrying the direction in which uncertainty moves. The asymmetry between the two event types is nonetheless respected in the empirical hierarchy: the introducing events are the cleaner analogue of the model's uncertainty shock, and the decomposition in Section 7 reports them separately rather than relying on the pooled signed set alone.

Secondly, MSR events arrive on a legislative calendar, whereas the benchmark model's τ is a single decision date. This is less restrictive than it appears. In their discussion of empirical implementation, Pástor and Veronesi (2012, Section VI, p. 1257) observe that most reforms occur gradually, clearing multiple hurdles, and that stock prices should respond at each step of the way, with larger responses at the steps that produce larger revisions in the perceived probability of the policy change; and Kelly et al. (2016) demonstrate that the framework applies to *scheduled* political events, where the date is known and only the outcome is uncertain. In the model's vocabulary, scheduled events carry no timing surprise but retain political uncertainty σ_c: the residual uncertainty about the decision's content at a known date. What the announcement resolves, and therefore what the announcement-window return can reflect is only this residual. The empirical implication is an attenuation result rather than an invalidation: to the extent that legislative outcomes were partially anticipated before the event window opens, the measured interaction is biased toward zero, and the estimated effect bounds the mechanism's realised magnitude from below. The pre-event window specification (-5 to -2 trading days) serves as the corresponding anticipation diagnostic. This attenuation logic is central to the interpretation of the results in Section 7: a tightly bounded null on the announcement window is a statement that the residual, unresolved component of MSR outcomes commanded at most an economically small repricing, not a statement that the discount-rate channel is absent.

Thirdly, the revision any single announcement can produce is scaled by the regime's expected remaining duration $(T - \tau)$ before the scheduled ETS review, and for revisions to policy uncertainty, the channel the clean-risk events isolate, the scaling is quadratic in the Pástor–Veronesi structure, where the posterior variance of the policy impact enters the log price in proportion to $(T - \tau)^2$. The shorter the horizon over which the MSR's rules are expected to bind, the more sharply compressed the announcement-window response the design measures; a null on late-phase events is accordingly the theory's own prediction, not a failure to detect.

One further feature of the mapping reinforces that the scope revolves around what survives attenuation. It concerns pooling. The premium for policy exposure is state-dependent in the model, as noted above, so a coefficient pooled over events spanning 2013–2025, from the reserve's novelty

to its maturity, averages event-specific responses that may differ in magnitude and, in principle, in sign. Partial anticipation and limited persistence shrink the measured effect toward zero; state dependence can additionally cancel across events. The estimated bound accordingly applies to the average carbon-scaled response across MSR events rather than to a single structural parameter. The decomposition into introducing and resolving subsets addresses the sharpest version of this concern: a null produced by opposite-signed subsets cancelling would surface as significant subset coefficients of opposite sign, and Section 7 shows both subsets are individually null.

4.5 Second moments: volatility, correlation, and design implications

The model's announcement effects are not confined to means. In Pástor and Veronesi (2012), the announcement is also a moment of elevated second moments: the volatility of the pricing kernel and of returns jumps at τ (their Propositions 6 and 7), and the pairwise correlation between stocks rises discretely, reaching one at the announcement instant itself, because the policy decision is a common shock (their Corollary 9). Under the heterogeneous-exposure structure of equation (4), this correlation is generated by differential loadings on a common factor. A date fixed effect removes the common mean shift on an event date but cannot remove differential loading: high-exposure firms continue to co-move in the residual, and most strongly on precisely the treated dates. This is the model-based rationale for two-way clustering on firm and date in the empirical specification: within-date cross-sectional dependence is not an incidental nuisance but a prediction of the framework, concentrated exactly where the treatment fires. The elevated return volatility on event dates likewise implies heteroskedasticity across treated and untreated dates, which the clustered errors accommodate.

The diffusion dynamics before the announcement carry a second implication. Pástor and Veronesi (2012, Section II.D) show that pre-announcement volatility is driven in part by fluctuations in the perceived probability of the decision: prices move as investors revise their views of what the government will do. Anticipation in the model is therefore primarily a second-moment phenomenon.

The pre-event window diagnostic in this thesis tests the *mean* of abnormal returns over the days preceding the announcement and accordingly detects directional anticipation only; belief revision that nets to zero in the mean leaves no trace in it. The diagnostic is read in that limited capacity in Section 7.

4.6 The competing cash-flow channel and separation by construction

The pricing identity behind equation (3) disciplines the design in one further way. MSR decisions that alter the allowance supply path move the expected cash flows of carbon-exposed firms directly: Bushnell et al. (2013) show that carbon-cost pass-through operates through merit-order electricity pricing regardless of the allocation regime, so a supply decision changes $E_t[B_T^i]$ for high-carbon firms even holding the kernel fixed. An event that simultaneously shifts regulatory uncertainty and the supply path therefore confounds the two channels, and, because the two effects can carry opposite cross-sectional signs, pooling such events risks cancellation that no signed regression can undo.

The design responds by separating the channels in the construction of the event set rather than in the estimation. The primary specification restricts to *supply-neutral* events, on which the cash-flow channel is inactive by construction and any carbon-scaled response identifies the risk channel alone; supply-active events enter only through diagnostic specifications that sign them by their cash-flow direction, providing the competing mechanism's own treatment as a placebo. In terms of the framework, the supply-neutral set holds $E_t[B_T^i]$ fixed across the carbon cross-section and lets only π_t move, while the supply placebo does the reverse. Hypothesis H3 formalises the resulting testable contrast, and Section 6 details the classification of events into these categories.

5 Data

This section describes the three inputs to the estimation: firm returns, carbon intensity, and the signed MSR event calendar and the sample each yields. Formal definitions of every constructed variable, with panel field names and sources, are collected in Appendix B.1.

5.1 Firm-Level Equity Data

Firm-level return data are collected from Compustat Global and validated against FactSet. Daily returns are used to capture short-term market reactions to regulatory information and are therefore chosen as opposed to the more established metric used in the literature, monthly returns.

The historical STOXX 600 is an unbalanced panel by design, so as to avoid survivorship bias. One thousand one hundred twenty-eight firms appear across the 2010–2025 period, against the 600 constituents held on any single day. After an approximately 48-month rolling-beta warm-up period (see the methodology section), 1,052 firms have at least one abnormal-return observation in the 2013–2025 window; the 76 firms lost at this stage enter the index too late, or trade too sporadically, to accumulate a full estimation window before exit. Of these 1,052 firms, 850 also carry the Refinitiv carbon-intensity data required for the interaction and constitute the final estimation sample. The further reduction of 202 firms therefore reflects carbon-data coverage rather than the return panel itself.

Because later index entrants cross their warm-up threshold only after roughly four years of history, the cross-section grows over the sample, from 832 firms in late 2013 to 876 by 2019, and averages 841 per trading day. The earliest events are thus identified on a slightly thinner cross-section than the later ones, which is a limitation of the design.

The sample reflects European equities potentially affected by the MSR mechanism. Some listed firms nonetheless fall outside the MSR's jurisdiction, which is a further limitation of the design.

Table 1 reports summary statistics for the two central variables. Panel A describes the daily abnormal

return AR_{it}, the six-factor rolling-beta residual that serves as the dependent variable, at the firm-day level and in basis points; its mean of -1.25 bp is economically negligible and consistent with a correctly centred residual. Panel B describes carbon intensity at the firm-year level over the 850 matched firms. The raw ratio is extremely right-skewed—spanning nine orders of magnitude, from 5.6×10^{-11} to 0.14, which motivates the within-year log standardisation: the resulting z-score is, by construction, close to mean zero and unit standard deviation.

5.2 Carbon Intensity Data

Firms are classified according to carbon intensity, measured as Scope 1 emissions scaled by revenue, both collected from the Refinitiv database:

$$\text{CarbonIntensity}_i = \frac{\text{Scope1Emissions}_i}{\text{Revenue}_i}. \tag{6}$$

The Refinitiv extract covers 1,106 firms over 2009–2025 in 24,725 rows. Firms for which either Scope-1 emissions or revenue is unreported are dropped, removing 221 firms and leaving 885; a further 32 rows carry non-positive revenue, and 115 rows report zero or negative Scope-1 emissions, which cannot be logged, removing five more firms. The 13,976 rows that remain are not distinct firm-years: the extract returns multiple rows for 1,428 firm-years, which are collapsed to one observation each, giving 8,442 firm-year observations across 880 firms. No observation is lost at that step, and none is lost at standardisation: every one of the 8,442 firm-years receives a within-year z-score.

Of the 880 firms, 851 match the abnormal-return panel on ISIN and 850 contribute at least one matched firm-day; those 850 account for 8,350 of the 8,442 standardised firm-years, the remaining 92 belonging to firms with coverage too thin to survive the match. Because the measure is lagged one year and carried forward until superseded, 6,898 of the 8,350 attach to at least one trading day inside the analysis window and thereby enter estimation; the remainder are disclosures from

2009–2012 that represent the period before the analysis window opens. The merged panel is 1,957,939 firm-days, 73.7% of the 2,657,470 in the return panel. This margin turns on identifiers and the calendar rather than on emissions: mean standardised carbon intensity is $+0.005$ among the firm-years that enter and -0.023 among those that do not, a gap of three hundredths of a standard deviation that is not statistically distinguishable from zero ($t = 0.98$, $p = 0.32$). The estimation sample is not selected on the regressor, so the null reported below cannot be read as an artefact of carbon-data coverage.

The raw ratio is unusable as a regressor in levels. Across the matched sample it spans nine orders of magnitude, from essentially zero for asset-light financial and services firms to roughly 1.4×10^{-1}(some 144,000 tonnes of CO_2 per revenue) for the heaviest utilities and materials producers. Its mean of about 258 tonnes per million sits more than twenty times above its median of 11.7, so the distribution is dominated by a small number of extreme emitters. Estimated in levels, the carbon term would therefore be identified almost entirely off a handful of firms, and a one-unit change in the regressor would have no interpretable meaning across the bulk of the cross-section. Figure 1 shows the distribution of carbon intensity divided by a million revenue across firm-years before and after the log transformation.

Taking natural logarithms converts this multiplicative dispersion into an approximately symmetric distribution, reported in Panel B of Table 1: the log mean of -11.17 and median of -11.36 are close, and the interquartile range of three log points corresponds to a roughly twenty-fold difference in emissions per unit of revenue between the upper and lower quartiles. The log variable is then standardised within each calendar year, so that firms are ranked against contemporaneous peers rather than against the sample as a whole. Within-year standardisation matters here because average reported intensity drifts downward over the sample, as both genuine decarbonisation and improving disclosure coverage change the composition of the reporting universe; pooled standardisation would confound that time trend with cross-sectional carbon exposure. Figure 2 portrays this transformation.

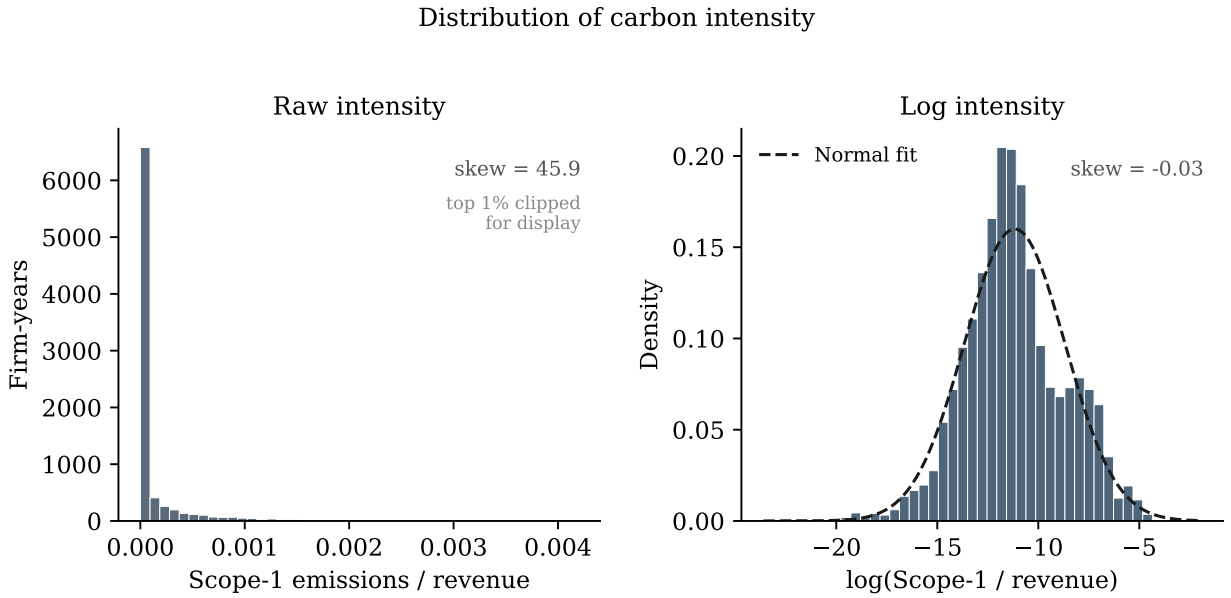


Figure 1: Distribution of carbon intensity across firm-years. The left panel shows the raw ratio of Scope-1 emissions to revenue, which is heavily right-skewed; the right panel shows $\log(\text{Scope-1}/\text{revenue})$, which is approximately symmetric, motivating the log transform before within-year standardisation.

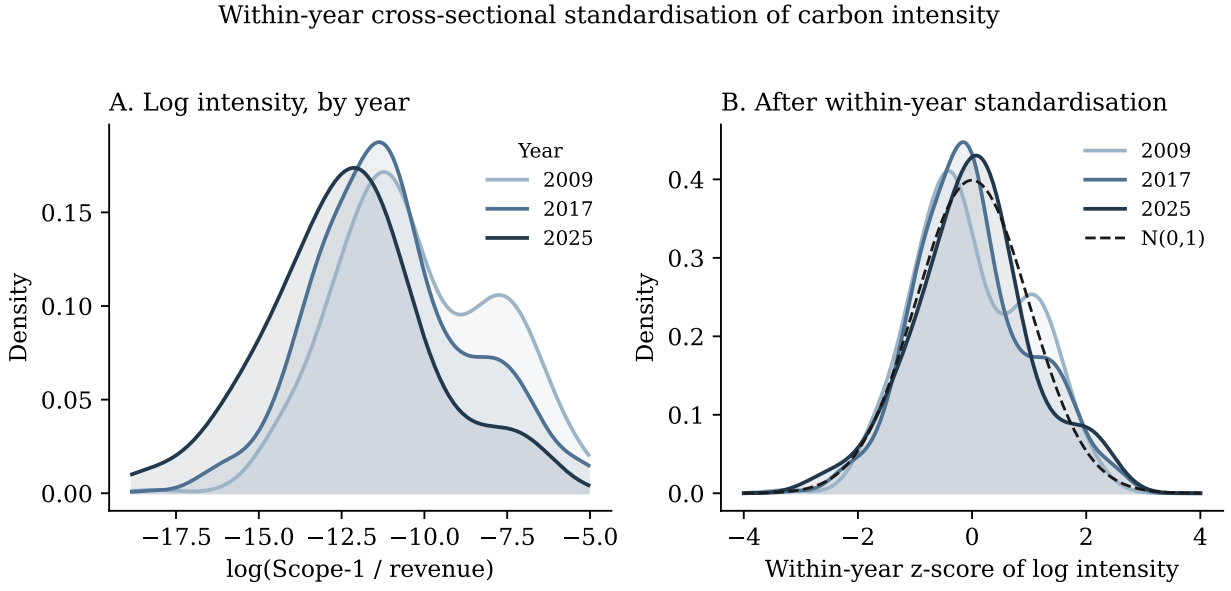


Figure 2: Within-year cross-sectional standardisation of carbon intensity. Panel A shows the distribution of $\log(\text{Scope-1}/\text{revenue})$ in three sample years; the cross-section shifts from year to year, so a firm's raw level does not fix its standing among peers. Panel B shows the same years after within-year standardisation, which collapses each year onto a common $\mathcal{N}(0,1)$, so that `carbon_intensity_z` measures a firm's position relative to its contemporaries.

The resulting variable, $\mathrm{carbon}_{z,it}$, has unit standard deviation by construction, so the interaction coefficients reported throughout the results section are read as basis points of daily abnormal return per one within-year standard deviation of log carbon intensity. Finally, the measure is lagged one year relative to the returns with which it is matched, so that the classification of a firm as high- or low-carbon on any event date uses only information that was already public.

Abnormal returns and carbon intensity are essentially uncorrelated in the pooled cross-section (pairwise correlation 0.002), as expected once firm fixed effects are absorbed. The identifying variation is not a raw carbon–return relationship but the interaction of carbon intensity with signed regulatory events, developed in the methodology section.

Table 1: Summary statistics

Variable	N	Mean	Median	SD	Min	P_{25}	P_{75}	Max
Panel A: Abnormal returns (firm-day)								
Abnormal return AR_{it} (bp)	1,957,939	-1.25	-1.12	219.66	$-9{,}814.67$	-89.11	86.93	9,992.88
Panel B: Carbon intensity (firm-year)								
log carbon intensity	8,350	-11.17	-11.36	2.49	-23.61	-12.70	-9.70	-1.94
Carbon intensity, z-score	8,350	0.00	-0.06	1.00	-4.76	-0.63	0.60	3.68

Notes. Panel A: AR_{it} is the Fama–French five-factor plus momentum residual from rolling 48-month betas lagged one day, expressed in basis points of daily return, over all firm-days in the final estimation panel.
Panel B: log carbon intensity is the natural logarithm of Scope-1 emissions divided by revenue (Refinitiv), at the firm-year level, restricted to the 850 firms that match into the return panel. The z-score standardises this variable within year on the full carbon universe before matching to returns, which is why its mean over the matched subset is near, but not exactly, zero. The interquartile range of 3.00 log points corresponds to a firm at the top of the middle half of the distribution emitting roughly twenty times as much per unit of revenue as a firm at the bottom.
N differs across panels by construction: AR is measured at the firm-day level and carbon intensity at the firm-year level. Carbon intensity is lagged one year relative to the returns with which it is matched.

5.3 Signed MSR Events

I assemble a list of MSR-related events over 2013–2025 following Borghesi et al. (2023), and cross-validate them.

Table 2 lists the full event set with its channel classification and the two sign codings that drive the specification ladder in the methodology section: the regulatory-risk sign s^{pv} and the supply

(cash-flow) sign s^{sup}.

The event set comprises all exact-dated MSR events falling inside the 2013–2025 estimation window, hand-classified from the legislative record and cross-validated against the news-intensity series. Each event carries two signs. The regulatory-risk sign s^{pv} is coded -1 for risk-introducing events and $+1$ for risk-resolving events, while the supply (cash-flow) sign s^{sup} is coded -1 where the event tightens expected allowance supply and $+1$ where it loosens it. These two dimensions define the event taxonomy used throughout the analysis. Clean-risk events are supply-neutral, carrying $s^{sup} = 0$, and unambiguous in risk direction, so their announcement response isolates the regulatory-risk channel. Confounded events carry non-zero signs on both channels; reinforcing events are the subset on which the two signs coincide, so both channels push high-carbon firms in the same direction; and supply-placebo events are risk-neutral ($s^{pv} = 0$) but supply-active, signed by s^{sup} in the placebo specification. Neutral events receive no event window and remain in the control pool.

A further five events—IDs 5, 7, 10, 12, and 13, all annual TNAC publications between 2017 and 2021—are dated only to the month, since the publication day could not be independently verified. Because the $[-1, +1]$ and $[-5, -2]$ windows require exact dating, these are excluded from the windowed specifications. The appendix (Table 7) lists all 27 events with full descriptions and legal references.

6 Empirical Strategy

6.1 Abnormal Return Estimation

Abnormal returns are the six-factor residual from a Fama–French five-factor model augmented with momentum. For firm i on day t, the daily USD excess return $R_{i,t} - R_{f,t}$ is modelled as

$$R_{i,t}-R_{f,t} = \alpha_i+\beta_{1i}\,(R_{m,t}-R_{f,t})+\beta_{2i}\,\mathrm{SMB}_t+\beta_{3i}\,\mathrm{HML}_t+\beta_{4i}\,\mathrm{RMW}_t+\beta_{5i}\,\mathrm{CMA}_t+\beta_{6i}\,\mathrm{MOM}_t+\varepsilon_{i,t}, \tag{7}$$

Table 2: MSR regulatory events, 2013–2025

ID	Date	Event	Analysis group	s^{pv}	s^{sup}
1	2014-01-22	Commission proposes the MSR	Reinforcing	-1	-1
2	2015-03-05	Council General Approach on MSR	Clean-risk	$+1$	0
3	2015-10-06	Decision (EU) 2015/1814 adopted	Confounded	$+1$	-1
4	2016-04-04	Poland files Case C-5/16 at the ECJ	Clean-risk	-1	0
6	2018-03-14	Directive (EU) 2018/410 adopted	Confounded	$+1$	-1
8	2018-06-21	ECJ judgment in Case C-5/16	Clean-risk	$+1$	0
9	2019-01-01	MSR becomes operational	Neutral	0	0
11	2019-12-11	European Green Deal Communication	Reinforcing	-1	-1
14	2021-07-14	Fit for 55 package – MSR proposal	Reinforcing	-1	-1
15	2022-05-13	TNAC 2021 publication	Neutral	0	0
16	2022-06-22	EP plenary vote on ETS revision fails	Clean-risk	-1	0
17	2022-06-29	Council General Approach on ETS revision	Clean-risk	$+1$	0
18	2022-12-18	Trilogue provisional agreement on ETS revision	Confounded	$+1$	-1
19	2023-01-01	First MSR invalidation ($\sim$2.5 bn allowances)	Supply placebo	0	-1
20	2023-04-19	Decision (EU) 2023/852 adopted	Confounded	$+1$	-1
21	2023-05-10	Directive (EU) 2023/959 adopted	Risk-ambiguous	$+1$	0
22	2023-05-15	TNAC 2022 publication	Neutral	0	0
23	2024-01-01	Second invalidation; revised rules take effect	Supply placebo	0	-1
24	2024-05-31	TNAC 2023 publication	Neutral	0	0
25	2025-01-01	Third annual invalidation	Supply placebo	0	-1
26	2025-05-28	TNAC 2024 publication	Neutral	0	0
27	2025-11-27	Commission proposes ETS2 MSR amendments	Confounded	-1	$+1$

Notes. All exact-dated MSR events falling inside the 2013–2025 estimation window, hand-classified from the legislative record and cross-validated against the news-intensity series. s^{pv} is the regulatory-risk sign, coded -1 for risk-introducing events and $+1$ for risk-resolving events; s^{sup} is the supply (cash-flow) sign, coded -1 where the event tightens expected allowance supply and $+1$ where it loosens it. Clean-risk events are supply-neutral ($s^{sup} = 0$). Confounded events carry non-zero signs on both channels; reinforcing events are those on which the two signs coincide; supply-placebo events are risk-neutral ($s^{pv} = 0$) but supply-active, and are signed by s^{sup} in the placebo specification. Neutral events receive no event window and remain in the control pool.

Five further events (IDs 5, 7, 10, 12 and 13, all annual TNAC publications between 2017 and 2021) are dated only to the month, since the publication day could not be independently verified; they are excluded from the $[-1, +1]$ and $[-5, -2]$ windows, which require exact dating. Table 7 in the Appendix lists all 27 events with full descriptions and legal references.

Table 3: Specification event sets

Specification	Signed by	# Events	Event IDs
Clean-risk (supply-neutral)	Risk	5	2, 4, 8, 16, 17
introducing	Risk	2	4, 16
resolving	Risk	3	2, 8, 17
Reinforcing	Risk	3	1, 11, 14
Supply placebo	Supply	3	19, 23, 25
Pooled signed	Risk	14	1, 2, 3, 4, 6, 8, 11, 14, 16, 17, 18, 20, 21, 27

Notes. "Signed by" indicates whether the treatment uses the regulatory-risk sign s^{pv} or the supply sign s^{sup}. Clean-risk is the primary identifying specification; its introducing and resolving cells decompose it by the direction of the risk shock. Reinforcing and supply-placebo events bound the competing cash-flow channel. Pooled signed conflates the two channels and is reported as a robustness check only.

using the European FF5 and momentum (MOM, winners-minus-losers) factors from the Kenneth French Data Library. The abnormal return is the realised excess return net of its factor-model expectation,

$$AR_{i,t} = (R_{i,t} - R_{f,t}) - \left(\hat{\alpha}_i + \textstyle\sum_{k=1}^{6} \hat{\beta}_{ki,\,t-1}\, f_{k,t}\right), \tag{8}$$

where $f_{k,t}$ collects the six factor realisations on day t and $(\hat{\alpha}_i, \hat{\beta}_{ki,\,t-1})$ are firm-specific loadings estimated through $t-1$.

Crucially, these loadings are not estimated on the full sample. A single full-sample fit would use 2013–2025 information to benchmark, say, a 2018 event, letting future data contaminate the expectation a market participant could have formed at the time. I therefore estimate the loadings on a trailing rolling window of 48 months ($1{,}008$ trading days) ending on day $t-1$, so every $AR_{i,t}$ is a surprise relative to a benchmark formed strictly out of sample. Lagging the estimated betas one day within each firm makes the construction transparent: over a window this long the influence of any single observation is negligible, but the lag removes any residual look-ahead objection.

Because factor loadings are a structural property of a firm rather than an artefact of the event sample, the estimator can be warmed up on data that are themselves unusable for the analysis. I estimate on 2010–2025 and trim the resulting AR series to 2013 onward: the 2010–2012 years, which are contaminated for analysis by Phase 2/3 allowance backloading, serve only to give the rolling window enough history for betas to be warm by the first event in January 2014. Firms that list later

warm up from their own entry date, so younger firms contribute fewer AR rows; a beta cannot be estimated without history, and this is the honest cost of the design rather than a selection problem. Days inside the warm-up window carry no estimate and drop out. The 48-month window is chosen over the 60-month alternative because the longer window's warm-up would push past the January 2014 event and cost a scarce observation; the 60-month specification is reported as a robustness check in Section 7.2.

6.2 Event Identification and Sign Coding

Identification rests on partitioning the MSR legislative events introduced in Section 5 by the channel through which they plausibly move equity prices, *before* any return is examined. Two channels are distinguished: a *regulatory-risk* (discount-rate) channel, through which news about the reserve's rules changes the policy uncertainty priced into carbon-exposed firms without altering the near-term allowance supply; and a *cash-flow* (supply) channel, through which news that changes the physical quantity of allowances moves expected compliance costs directly. Events are classified as *supply-neutral* or *supply-active* on this basis. This ex-ante partition is the central methodological choice of the design: rather than attributing a single pooled price reaction to competing channels ex post as a Campbell-Shiller decomposition does (Ardia et al., 2023) the partition separates the channels by construction, so each subsample speaks to one channel by design.

Within the supply-neutral set, events are signed by the direction of the regulatory-risk shock, $s_t = \texttt{pv_ar_sign} \in \{-1, 0, +1\}$: -1 on risk-*introducing* days, on which a new source of policy uncertainty enters the price; $+1$ on risk-*resolving* days, on which uncertainty is retired; and 0 outside the event windows. The introducing sign is anchored directly in the announcement model of Pástor and Veronesi (2012), which predicts a first-moment price reaction to the arrival of policy news; the resolving sign is the mirror-image extension of the same logic, with indirect support from the scheduled-event framework of Kelly et al. (2016). Supply-active events carry a second sign, `supply_ar_sign`, giving the cash-flow direction of each event; this signing is used only in the

channel-separation tests of Section 6.4.

Two event windows are used throughout. The announcement window $[-1, +1]$ around each event date captures the repricing attributable to the news. The pre-event window $[-5, -2]$ is a diffusion/placebo window: under gradual anticipation, part of any reaction should appear here, so a null in this window is evidence that the announcement-window signal, if any, is not merely leakage.

6.3 Panel Event-Study Specification

The panel regression stacks all firm-days and interacts the signed shock with carbon intensity:

$$AR_{i,t} = \beta \left(s_t \cdot z_{i,t} \right) + \gamma \, z_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}, \tag{9}$$

where $z_{i,t}$ is carbon intensity, the within-year cross-sectional z-score of $\ln(\text{Scope 1}/\text{Revenue})$, lagged one year (Section 5), α_i and δ_t are firm and day fixed effects, and standard errors are two-way clustered by firm and date. The coefficient of interest is β, the carbon-amplification of the signed regulatory-risk shock, expressed in basis points of daily abnormal return per one within-year standard deviation of carbon intensity. Every variable entering the estimation is defined, with its panel field and source, in Table 6).

The day fixed effect δ_t absorbs every market-wide shock on an event date, including the level of the factors themselves, so β is identified purely from the cross-sectional gradient in AR across firms of differing carbon intensity on the same day. A positive β places high-carbon firms in the amplification direction: they react more strongly to regulatory-risk news than low-carbon firms do, in the sign the discount-rate channel predicts. I read this cross-sectional amplification as an operationalisation of the Pástor–Veronesi framework by the applied climate-finance literature (Hengge et al., 2023; Ramelli et al., 2021), not as a formal proposition of the original model. The model licenses a first-moment reaction to policy news (Pástor & Veronesi, 2012) and a discount-rate premium (Pástor & Veronesi, 2013); the further claim that this reaction loads on firm-level carbon intensity is the

applied literature's extension, and it is that extension the interaction term tests.

6.4 Channel-Separation Strategy

Three specifications map the two channels onto the data. The *clean-risk* (primary) specification restricts to the supply-neutral events signed by `pv_ar_sign`, isolating the discount-rate channel. Within it, the introducing-only subset is the cleanest identification: the risk-resolving $+1$ sign is the fragile assumption, because some resolving events bundle premium compression with directional tightening content, and pooling potentially mis-signed resolving events can reintroduce return cancellation even in a signed regression. The *supply-placebo* specification takes the supply-active events and signs them by `supply_ar_sign`, the competing channel's own treatment, so that a cash-flow effect, if present, should surface here even where the risk signing detects nothing. The *reinforcing* specification is the subset of events on which the risk and supply signs coincide, signed by `pv_ar_sign`: the maximum-signal case, in which both channels push high-carbon firms the same way.

The three specifications combine into a single inference. A clean-risk null pins the pure regulatory-risk effect near zero. Given that null, a reinforcing null then forces the cash-flow effect near zero as well. Even when both channels are aligned to push in the same direction, nothing appears. Finally, the supply-placebo null paired with the risk-sign null is the strongest rebuttal to an underpowered-null reading: the design detects no carbon-scaled response under *either* channel's own signing, which is difficult to reconcile with a large true effect masked by low power.

6.5 Inference

Two-way clustering by firm and date is appropriate for the pooled clean-risk specification, which spans enough distinct event dates to support date clustering. The introducing, resolving, supply-placebo, and reinforcing subsets each span at most three same-signed event dates, too few for

reliable date clustering, so these rows are firm-clustered and reported as point estimates against their confidence intervals rather than as significance tests. Firm-only clustering understates the within-date cross-sectional correlation of high-carbon firms' abnormal returns, so its standard errors are read as a lower bound.

Several features of the setting shrink the announcement-window response relative to the structural regulatory-risk effect, so a null coefficient is interpreted throughout as a bound on the residual repricing that reaches the event window, not as evidence that regulatory risk is unpriced; this reading is developed in Section 7.3.

7 Empirical Results

7.1 Main Results

Panel A of Table 4 reports the primary result. In the pooled clean-risk specification the carbon-scaled announcement-window coefficient is 0.80 bp per within-year standard deviation of log Scope-1 carbon intensity (SE 4.37), indistinguishable from zero: high- and low-carbon firms reprice by statistically the same amount in the three-day window around supply-neutral regulatory-risk events. As such, **H2 is not supported.** The content of this null lies in its precision. The 95% confidence interval for this pooled, two-way-clustered coefficient reaches only 9.4 bp from zero ($|\hat{\beta}| + 1.96\,\mathrm{se}(\hat{\beta})$; the computation is documented in Appendix C.1), so a carbon amplification larger than that magnitude is rejected by the observed data at the 5% level. Regulatory-risk news on supply-neutral MSR events moves high-carbon firms by at most single-digit basis points more than low-carbon firms, per standard deviation of carbon intensity; the pure regulatory-risk channel is bounded near zero rather than left undetermined.

The signed specification already protects this pooled estimate from averaging a genuine directional effect toward zero: introducing and resolving events are coded so that amplification accumulates in one direction rather than cancelling. That protection is only as good as the signs, and the resolving

Table 4: Carbon-scaled abnormal-return response to signed EU ETS Market Stability Reserve events

Specification	Cluster	Events	Event window	
			Announcement $[-1,+1]$	Pre-event $[-5,-2]$
Panel A. Regulatory-risk channel: supply-neutral events				
Clean-risk (all)	2-way	5	0.80 (4.37)	-5.38^{*} (3.16)
Introducing	firm	2	4.61 (3.22)	-0.10 (2.66)
Resolving	firm	3	-1.83 (2.80)	$-9.02^{\dagger}$ (2.04)
Panel B. Channel separation: supply-active events				
Supply placebo	firm	3	3.42 (3.37)	1.71 (1.09)
Reinforcing	firm	3	-0.60 (2.82)	0.47 (1.83)

Notes. Variable definitions, panel field names, and sources are in Table 6; the event classification is in Table 7.
Stars are assigned to two-way (firm + date) clustered rows only: $^{*}\,p<0.10$, $^{**}\,p<0.05$, $^{***}\,p<0.01$.
† Not statistically significant under two-way (firm + date) clustering; see Section 7.
Standard errors in parentheses.

$+1$ coding is the design's most fragile assumption, the danger is that of a mis-signed resolving set could reintroduce the very cancellation the signing is meant to prevent, even within the signed regression. The decomposition into sign-homogeneous halves is what lets the pooled null be read accurately against this risk: not a discovery that the null is clean, but a check on whether it is genuine or a signed-but-cancelling artefact. It is genuine. The introducing events, whose risk sign is unambiguous and which therefore give the cleanest identification, carry 4.61 bp (SE 3.22): positive, in the Pástor–Veronesi amplification direction, but with a confidence interval that comfortably includes zero. The resolving events return -1.83 bp (SE 2.80), near zero and if anything weakly against the amplification prediction. These are not the large, offsetting pair a masking mis-sign would produce; their event-weighted combination closely reproduces the pooled 0.80, so pooling mildly attenuates the introducing estimate rather than cancelling a real effect. The introducing-only figure is thus the cleaner read, and even it does not clear its own interval. The two subsets decompose the pooled estimate rather than testing it independently, and both are firm-clustered across at most three same-signed dates, so their coefficients are read as point estimates against their intervals, not as significance tests.

The pre-event window guards against the possibility that an announcement-window null merely reflects news that had already diffused into prices. The pooled clean-risk pre-event coefficient is -5.38 bp and marginally significant (SE 3.16, $p < 0.10$), which in isolation would suggest anticipatory leakage. It does not survive scrutiny. The signal is carried entirely by the resolving subset (-9.02), while the sign-unambiguous introducing events show no pre-event response whatsoever (-0.10, SE 2.66). The resolving figure is itself a clustering artefact. Firm-only clustering treats the abnormal returns of different high-carbon firms as independent within an event date; that within-date correlation is in fact positive, common date-level shocks that load on carbon, such as an energy-price move, push high-carbon firms in the same direction, so setting it to zero understates the standard error, and across only three event dates the understatement is severe.[1] Under two-way (firm + date) clustering the coefficient is not statistically significant. It is moreover wrong-signed for the resolution hypothesis; resolving events are coded $+1$, so amplification predicts a positive coefficient and does not survive narrowing the event window. There is therefore no genuine pre-event anomaly on the events where the risk sign is clean, and the announcement-window null cannot be attributed to leakage.

Panel B bounds the competing cash-flow channel. The supply-placebo row re-signs the supply-active events by their cash-flow direction (`supply_ar_sign`), the competing mechanism's own treatment, under which a genuine cash-flow effect should surface, and returns 3.42 bp (SE 3.37): a small point estimate, well below the announcement bound and consistent with no cash-flow effect. **H3 is not supported.** The reinforcing row, the subset of events on which the risk and supply signs coincide so that both channels push high-carbon firms the same way, the maximum-signal case, it returns -0.60 bp (SE 2.82), a still tighter near-zero estimate. Neither pre-event coefficient (1.71 and 0.47) shows anticipatory movement. Both rows are firm-clustered across three event dates and are accordingly read as point estimates rather than significance tests; for the same reason they do

[1] The day fixed effect δ_t removes the *level* of such a common shock, but not its differential loading on carbon intensity, which is the component correlated with the regressor. Fixed effects control the mean of the within-date dependence; clustering corrects the variance for what remains. The two are distinct corrections, which is why δ_t does not make date clustering redundant.

not enter the computation of the effect bound (Appendix C.1).

Read together, the two panels close the obvious escape routes for a hidden effect. The clean-risk null places the residual regulatory-risk effect at approximately zero. Given that null, the reinforcing null forces the cash-flow effect near zero as well: when both channels are aligned to move high-carbon firms in the same direction, the response is still nothing. Since the supply-placebo events are null under their *own* cash-flow signing, not merely under the risk signing, the result cannot be dismissed as a mis-specified treatment; the cash-flow mechanism fails to surface even when handed its best-case sign. The pooled two-way-clustered specification bounds the carbon-scaled announcement response at 9.4 bp per standard deviation of carbon intensity, roughly ± 10 bp in round terms, and the firm-clustered subsets all lie below 5 bp in magnitude.

Finally, **H1 is not supported.** Pooling all fourteen signed MSR events, the carbon-scaled announcement-window response is 0.58 bp (SE 2.45), 95% CI $[-4.2,\ 5.4]$ bp. This specification is reported in Table 5 rather than among the primary results because pooling conflates the regulatory-risk and cash-flow channels, which the design is built to separate ex ante; it is nonetheless the direct test of the baseline amplification claim, and it is the tightest of the three bounds.

To conclude, none of the three hypotheses is supported. The estimates bound the carbon-scaled response to within ± 10 bp per standard deviation in the primary specifications. The informativeness of this result rests on the ex-ante channel separation: because the treatment distinguishes regulatory-risk from supply-active events before estimation, the H2 null identifies the discount-rate channel specifically, and the H3 null then closes off the cash-flow explanation rather than leaving it as an unexamined alternative.

Section 7.3 sets out what this bound does and does not license: the forces that shrink announcement-window repricing, such as anticipation under a forecastable rule, the regime's finite expected duration, and attenuation from measurement error and sign misclassification, imply the structural regulatory-risk effect is weakly larger than the residual measured here, so the null bounds the repricing that survives anticipation, not the existence of priced regulatory risk.

7.2 Robustness Checks

The announcement-window null is stress-tested along five dimensions. *Pooling all signed events* replaces the supply-neutral set with the full set of signed MSR events; because this conflates the regulatory-risk and cash-flow channels, it is a robustness check rather than the primary specification. The *60-month beta window* re-estimates the primary specification with a longer rolling window, reconstructing AR upstream (Section 6.1) and confirming the null is not an artefact of window length. The *quartile carbon* check replaces the continuous z-score with a top-minus-bottom carbon-quartile contrast; its coefficient is scaled per quartile rather than per standard deviation, so its sign and confidence interval, not its magnitude, are the object of interest. The *energy-return* $\times$ *carbon* control adds an interaction of the daily Brent crude return (FRED series `DCOILBRENTEU`; see Table 6) with $z_{i,t}$ to absorb the one confound the day fixed effect cannot: time-varying energy-price moves that hit high-carbon firms asymmetrically on event dates. The EU allowance price is deliberately not used as this control, since it lies on the causal path of the events under study. Finally, following Aswani et al. (2024), the specification is re-estimated on firm-*reported* emissions only, to guard against measurement artefacts in vendor-estimated emissions data; because that check re-runs every row of the specification ladder rather than the primary specification alone, it is reported separately in Table 8 in the appendix and discussed at the end of this subsection.

Across all four perturbations the announcement-window coefficient remains statistically indistinguishable from zero and with the single, anticipated exception of the quartile contrast, within one basis point of the ± 10 bp band, established by the primary specification: the widest interval, from the 60-month beta window, reaches 10.3 bp. The point estimate ranges from 0.58 to 1.60 bp per standard deviation across the continuous-measure rows, none significant at conventional levels: the null is not an artefact of the beta-window length, of the energy-price confound the day fixed effect cannot absorb, or of the sharp subsample's limited power. That last point is worth isolating. Pooling all fourteen signed events tightens the standard error to 2.45 bp which is the most precise announcement estimate in the study. Nevertheless, the coefficient (0.58 bp) still sits on zero, so

Table 5: Robustness of the carbon-scaled null

Robustness check	Cluster	Events	Event window	
			Announcement $[-1,+1]$	Pre-event $[-5,-2]$
Memo: clean-risk baseline[a]	2-way	5	0.80 (4.37)	−5.38* (3.16)
Pool all signed events	2-way	14	0.58 (2.45)	−0.45 (2.11)
Variations on the primary (clean-risk) specification:				
60-month beta window	2-way	5	1.60 (4.43)	−5.77* (3.32)
Quartile carbon (Q4−Q1)[b]	2-way	5	4.11 (12.35)	−12.55 (9.75)
Energy return × carbon control	2-way	5	0.75 (4.26)	−5.79* (2.98)

Notes. Coefficient on sign × carbon$_z$ (basis points per within-year standard deviation of log Scope-1 carbon intensity), from the specification in Table 4.

[a] Reproduced from Panel A of Table 4 for comparison only; it is the specification the rows below perturb, not a robustness check.

[b] The quartile coefficient is scaled per top-minus-bottom carbon quartile, not per standard deviation; its magnitude is not directly comparable to the other rows.

Standard errors in parentheses; stars as in Table 4 (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

the clean-risk null cannot be dismissed as a failure to reject driven by too few event dates. The quartile check is the one row whose interval exceeds the ± 10 bp bound, and does so by construction: replacing the continuous z-score with a top-minus-bottom quartile indicator discards within-quartile variation and rescales the coefficient per quartile rather than per standard deviation, inflating its standard error to roughly three times that of the continuous-measure rows. Its sign is positive as per the Pastor–Veronesi–predicted amplification direction and it does not flip, but its magnitude is not comparable across rows and its confidence interval, not its point estimate is the object of interest.

The pre-event column is reported for completeness rather than as a second test. The marginally significant clean-risk pre-event coefficient (-5.38 bp) reproduces almost unchanged under the longer beta window (-5.77) and the energy control (-5.79), but collapses toward zero once the event set is pooled (-0.45). This is exactly the pattern the channel decomposition in Table 4 implies: the pre-event signal originates in the resolving subset, where it is a firm-clustering artefact across three event dates, absent from the sign-unambiguous introducing events, and it is diluted, not amplified, by adding events from outside the clean-risk set. Its survival across window length and energy exposure confirms only that neither is its source; it does not reinstate a genuine risk-channel

anticipation effect, which the introducing events rule out.

The announcement-window null, by contrast, is stable in sign, magnitude, and precision across every perturbation. Taken together, the robustness battery does not merely fail to overturn the null, but it shows the null is insensitive to the discretionary choices in the design, so the ± 10 bp figure is a stable bound on the *residual* announcement-window repricing the carbon-scaled regulatory-risk effect that survives anticipation rather than an artefact of any single specification or of insufficient power. The bound is scoped to that residual object by design. Because partial anticipation, proxy measurement error, and sign misclassification each shrink the estimate toward zero, the measured effect understates whatever unattenuated structural premium may exist; the analysis therefore claims only what it identifies: the repricing observable in the announcement window and does not read ± 10 bp as a bound on the structural effect the attenuating forces would partly obscure.

Data provenance: company-reported versus vendor-estimated emissions. A final robustness concern targets the emissions inputs rather than the event design. Aswani et al. (2024) document that a large share of vendor-supplied carbon figures are not company-disclosed but back-filled by proprietary estimation models that are largely deterministic functions of firm size and sector, so that a carbon–return relationship estimated on such data can be mechanical rather than economic. In the present setting the concern runs in the opposite direction: because the headline result is a null, the operative question is not whether estimated data *manufacture* a spurious effect but whether estimation noise in `carbon_intensity_z` *attenuates* a real one. I therefore re-estimate every specification on the subset of firm-years whose Scope-1 value is company-reported, using the Refinitiv `CO2EstimationMethod` flag to drop the vendor-modelled observations (the *Median*, *CO2*, and *Energy* estimation models). The restriction removes roughly 2.5% of carbon-scaled firm-day observations.

The result is reported in Table 8. Both two-way-clustered announcement-window bounds remain inside ± 10 bp, and the identifying clean-risk estimate is essentially unmoved ($0.80 \rightarrow 0.95$ bp, bound 9.9 bp); the firm-clustered rows stay below 6 bp as point estimates. The pre-event structure

that motivates the footnote to Table 4 reproduces exactly on reported data: the sign-unambiguous introducing events remain null ($-0.10 \rightarrow +0.25$), while the pooled clean-risk pre-event signal continues to be carried entirely by the wrong-signed, firm-clustered resolving subset ($-9.02 \rightarrow -9.35$). The channel-separation logic thus does not depend on retaining vendor-estimated firm-years. Notably, all six announcement-window estimates move marginally *away* from zero under the restriction which is the direction predicted if vendor estimation injects classical measurement error into `carbon_intensity_z` and thereby attenuates the interaction toward zero. The moves are sub-basis-point and partly reflect a change in cross-sectional composition, so I read them as consistent with, rather than confirmation of, the attenuation channel; the substantive conclusion is that the null survives on the cleanest available inputs. The one estimate that sharpens materially, introducing-only ($4.61 \rightarrow 5.54$ bp, $p = 0.09$), is firm-clustered across two event dates with a confidence interval that still spans zero, and is not interpreted as a significant effect.

7.3 Reading the Null: What the Bound Does and Does Not License

The estimated null is informative rather than empty, but its information must be attached to the right object. The two-way-clustered specifications bound the carbon-scaled repricing realised in the announcement window within roughly ± 10 basis points per standard deviation of carbon intensity (9.4 bp at the interval endpoint; Appendix C.1), and the firm-clustered subsets all lie below 5 bp in magnitude. What the interval bounds, precisely, is the *residual announcement-window repricing*, the surprise component of a typical sample event as it reached prices in the three-day window, scaled by measured carbon intensity. For that object, the bound is exact.

Four features of the setting shrink this measured object relative to the structural regulatory-risk effect, and the direction of the shrinkage matters: each force pushes the estimate toward zero, so the structural effect is weakly *larger* in magnitude than what the window records. The confidence interval therefore does not bound the structural effect at ± 10 bp; a bound on any deeper object inherits the estimated bound divided by an attenuation product that the data do not identify, and

loosens accordingly (Appendix C.2 makes this translation explicit under calibrated scenarios).

Two of the forces operate on the estimand itself and are economic rather than econometric. Under *partial anticipation*, much of each event's content diffuses into prices before the window opens. Since the MSR is a forecastable rule, its legislative milestones scheduled and its mechanics deterministic, the window captures only the residual surprise, and the event indicator fires at full weight regardless of how much of the news was already priced. Under *regime persistence*, the revision any single announcement can deliver is scaled by the effective horizon $(T - \tau)$ over which the regime is expected to remain in force before the scheduled ETS review; a finite expected duration compresses the revision even a fully surprising event could produce.[2] Neither force hides a large announcement effect: under this mechanism the true announcement-window response is genuinely small. For this pair the null is therefore not weakened but reinterpreted; it is evidence about how the regime is priced, gradually and ahead of events, rather than about whether it is priced.

The other two forces are econometric attenuators in the classical sense, and for these the honest statement is that the bound on the true announcement-window gradient is looser than the estimated one by an unknown attenuation factor. *Measurement error* in carbon intensity biases the loading on the mismeasured regressor toward zero, and *sign misclassification* (concentrated in the risk-resolving $+1$ coding, whose events can bundle premium compression with directional tightening content) partially cancels same-window responses even inside the signed regression. The design answers each internally rather than by assumption. The introducing-only subset removes sign misclassification by construction, since the introducing -1 coding is unambiguous, and its coefficient still fails to clear its own interval. The robustness re-estimation on firm-reported emissions restricts to the observations where vendor measurement error is smallest, and the quartile-contrast specification is insensitive to error within quartile. The two attenuators that could loosen the bound are thus the two the design was built to check.

What the results license is accordingly scoped. They rule out carbon-scaled repricing of regulatory-

[2]For revisions to policy *uncertainty*—the channel at issue for risk-introducing and risk-resolving events—the compression is stronger still: in the Pástor–Veronesi log-price structure the posterior variance of the policy impact enters in proportion to the *squared* remaining horizon, so uncertainty revisions reprice with $(T - \tau)^2$ rather than $(T - \tau)$.

risk news larger than roughly ± 10 bp per standard deviation in the announcement window: per event of typical sample magnitude, the place where, under the standard event-study logic, a priced shock should leave its mark. They do not rule out a priced regulatory-risk premium incorporated gradually outside announcement windows; indeed, for a forecastable rule that is what the anticipation mechanism predicts, and the absence of pre-event drift on the sign-unambiguous events is consistent with incorporation that is smooth rather than clustered just ahead of announcements. The null is a statement about the channel through which MSR regulatory risk reaches prices, not a verdict on whether it is priced at all.

8 Policy Implications and Directions for Further Research

8.1 Policy implications

The premise motivating this study is Bushnell, Chong and Mansur's: how equity markets absorb carbon regulation is itself an input to setting that regulation's level and timing (Bushnell et al., 2013). The finding here is that a rule-based, formulaic quantity mechanism produced no detectable *residual* cross-sectional repricing at announcement: the carbon-scaled announcement-window effect is bounded within roughly ± 10 basis points per standard deviation of carbon intensity. The interpretation must respect what an event window measures: it captures only the repricing left to occur when residual outcome uncertainty resolves, not any repricing that had already been incorporated as the mechanism's rules-based trajectory became forecastable. A null is therefore the signature one would expect of a predictable rule, its effects diffuse into prices ahead of any single event rather than concentrating at it, rather than evidence that carbon regulation carries no premium. Read for policy, this bears on the *form* of the market's response: transparent, mechanical adjustment does not generate the abrupt, carbon-differentiated repricing shocks that discretionary or surprise interventions could, precisely because a rule is anticipable. The scope is otherwise narrow. This concerns cross-sectional *equity* repricing by carbon intensity, not the allowance price,

which does respond to supply news (Känzig, 2023); and, as Section 7.3 develops, the absence of an announcement response is consistent with gradual incorporation and does not establish the absence of a cost. The supply-placebo result speaks directly to a live legislative question. On 1 April 2026 the Commission proposed ceasing the invalidation of reserve allowances held above the 400-million threshold, retaining them instead as a buffer (European Commission, 2026a). The events on which this study's placebo is built are announcements about precisely that permanent-cancellation mechanism. That these events did not differentially move high-carbon equity values, even when signed by their cash-flow direction rather than their regulatory-risk direction, implies that the equity-market cost of retaining versus abolishing invalidation is second order. This is consistent with the Commission's own revealed judgement that the mechanism's task, absorbing the historical surplus, is substantially complete: by the end of 2024 some 3.2 billion allowances had been invalidated. Here the thesis stops being retrospective and becomes evidence bearing on an open decision.

Two cautions temper any policy reading of the null. First, on causality: the MSR events sit within a monotone tightening trend, so a single announcement carries only limited marginal information: investors price the direction of travel, not each individual step. This is the policy-facing form of the anticipation argument: the absence of announcement-window repricing must not be read as the absence of a regulatory cost, and event-study evidence should inform level-and-timing decisions only with that caveat attached. Second, a boomerang logic applies: if retrenchment today raises the expected stringency of future policy (Ramelli et al., 2021), then risk-*resolving* events are theoretically ambiguous in sign, because a softening announcement can raise rather than lower the expected future burden on high-carbon firms. This same mechanism explains, ex post, why the resolving subset is sign-fragile while the sign-unambiguous introducing events are not.

There is a natural economic reason a cash-flow shock might spare equity holders. Under merit-order pricing, the carbon cost is embedded in the electricity price regardless of whether allowances are auctioned or freely allocated, so the burden is passed through downstream rather than borne by shareholders (Bushnell et al., 2013). The equity-side null is therefore consistent with a distributional

picture in which the incidence of carbon regulation falls on consumers and product markets rather than on the owners of carbon-intensive firms, an implication that a shareholder-value lens alone would miss, and one that bears on the fairness case the Commission's accompanying Social Climate Fund is designed to address.

8.2 Directions for further research

The most immediate extension is not hypothetical: the experiment is being re-run. The invalidation mechanism whose introduction furnished this study's placebo events is the very mechanism now being wound back (European Commission, 2026a), within a broader Phase-5 revision of the ETS tabled on 17 July 2026 (European Commission, 2026b). This yields an almost literal out-of-sample test: where this thesis observed how the cross-section responded as permanent cancellation was *introduced*, the current cycle offers the symmetric observation as it is *removed*. The signed-event framework and the clean-risk partition apply to that cycle unchanged, and the events are already being written into the legislative calendar.

Three methodological avenues would sharpen identification beyond what this design supports. High-frequency identification around announcement timestamps, using intraday EUA-futures movements as a first stage in the manner of Känzig (2023), would address the anticipation problem directly rather than bounding it. Options-implied tail measures (Ilhan, Sautner, & Vilkov, 2021) would test whether higher moments move on event dates even where the conditional mean does not, a natural next question given a mean-zero result. Credit markets, where regulatory risk should bind most tightly for the most carbon-intensive issuers, offer a second asset class in which the same events can be priced.

9 Conclusion

This thesis asked whether the regulatory risk carried by the EU Emissions Trading System's Market Stability Reserve is priced in the cross-section of European equity returns, and whether carbon intensity is the margin along which that risk loads. To answer it, the design partitions the MSR event calendar into supply-neutral events, which shift the regulatory-risk environment without moving the allowance supply path, and supply-active events, which do both. Signing the supply-neutral set by the direction in which regulatory-risk uncertainty is introduced or resolved isolates a discount-rate channel (Pástor & Veronesi, 2012, 2013) from the competing cash-flow channel that operates through allowance scarcity (Bushnell et al., 2013) *by construction* rather than by statistical attribution. This is the study's central methodological contribution: where Ardia et al. (2023) separate the two channels ex post through a return decomposition, the partition here separates them ex ante through the event taxonomy itself, so that the risk-channel estimate is never contaminated by supply news it was not designed to capture.

The central finding is a precisely estimated null. Across every main specification the carbon-scaled response to supply-neutral regulatory-risk events is statistically indistinguishable from zero, and the 95% confidence interval bounds it to within roughly ± 10 basis points of daily abnormal return per within-year standard deviation of log Scope-1 carbon intensity. First-order repricing of carbon regulatory risk along the MSR event calendar can be ruled out at that magnitude. The single marginal signal, represented by a pre-event coefficient in the pooled clean-risk window, localizes entirely to the resolving subset, is absent among the sign-unambiguous introducing events, and does not survive two-way clustering; it is a clustering artefact, not a repricing anomaly. The channel-separation diagnostics extend rather than qualify this null: the supply placebo and the reinforcing subset leave the cash-flow channel bounded near zero as well, so the result is not an artefact of having examined only the risk margin. Because MSR outcomes emerge from a drawn-out legislative calendar, partial anticipation attenuates any true effect toward zero. In a literature where carbon-premium estimates change sign across measurement and specification choices (Pástor et al., 2022; Zhang, 2025), a

tightly bounded zero is itself information: a restriction on the set of models the data will support.

Finally, the experiment this thesis runs is not closed. The MSR is being redesigned as this is written: European Commission (2026a) sets out a package of structural reforms to the reserve, and the review proposal of 17 July 2026 (European Commission, 2026b) carries them further ahead of the scheduled ETS review. The permanent-invalidation mechanism whose announcements furnish this study's placebo events, the puncture of the waterbed that rendered post-2018 supply shocks non-neutral, is itself now slated for revision, so the very events used here to bound the cash-flow channel describe a regime on its way out. The signed-event framework developed in this thesis, and the clean-risk partition in particular, applies directly to the regulatory cycle now underway: the next round of MSR events is already being written into the legislative calendar, and the design built here is the instrument for pricing it. The next empirical evidence is already being recorded.

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A Documentation of AI Usage

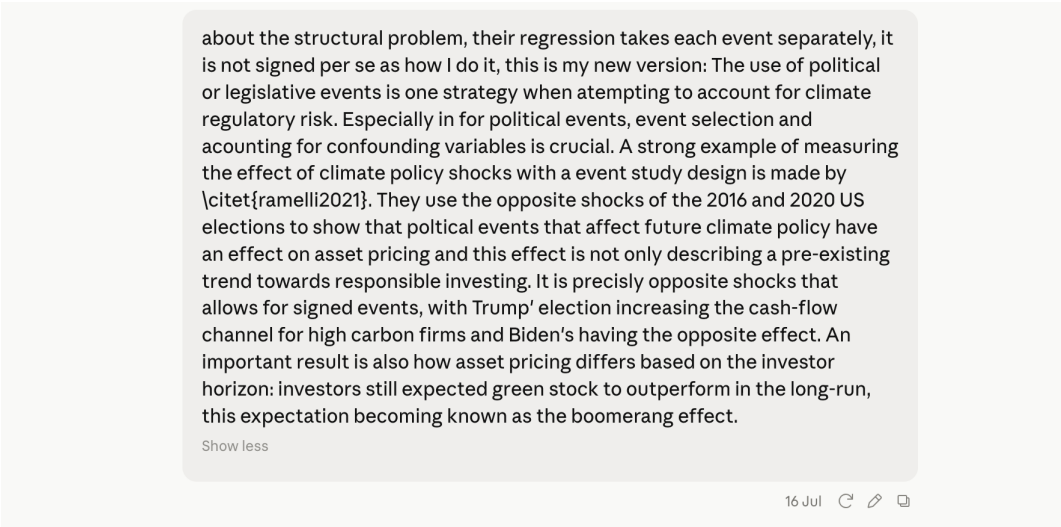
This appendix documents the use of a large language model (Anthropic's Claude) during the thesis process, in accordance with the statement of originality and the program's guidelines on responsible AI usage. The model was used as an interlocutor and critical reader: for pressure-testing methodological choices, sharpening the framing of the literature review, and checking the internal consistency of arguments.

Each example below records a genuine exchange: the prompt, a summary of the model's response, and an account of why the suggestion was adopted, modified, or rejected. The examples are chosen to show the points at which engaging with the model changed, corrected, or was overruled by the author's own reasoning.

Example 1: Positioning the closest structural precedent

During the literature review, the paragraph situating Ramelli et al. (2021) as the closest structural precedent for a signed climate-policy event study went through several rounds of revision with the model. The exchange reproduced below is the turn at which the model's proposed framing was overruled.

Example 1: Positioning the closest structural precedent



Key prompt (a follow-up turn): rejecting the model's characterisation of Ramelli et al. (2021) and supplying a corrected passage.

In the preceding turns the model had done two things. It flagged a citation risk, noting that the evidence on analyst forecast revisions might belong to the companion paper by Wagner, Zeckhauser and Ziegler (2018) rather than to Ramelli et al. (2021), and advised confirming which paper carried the cash-flow evidence before citing it. It proposed a condensed rewrite of the paragraph that presented Ramelli et al.'s design as a signed political-event study interacted with cross-sectional climate exposure describing it, in effect, as the same specification used in this thesis.

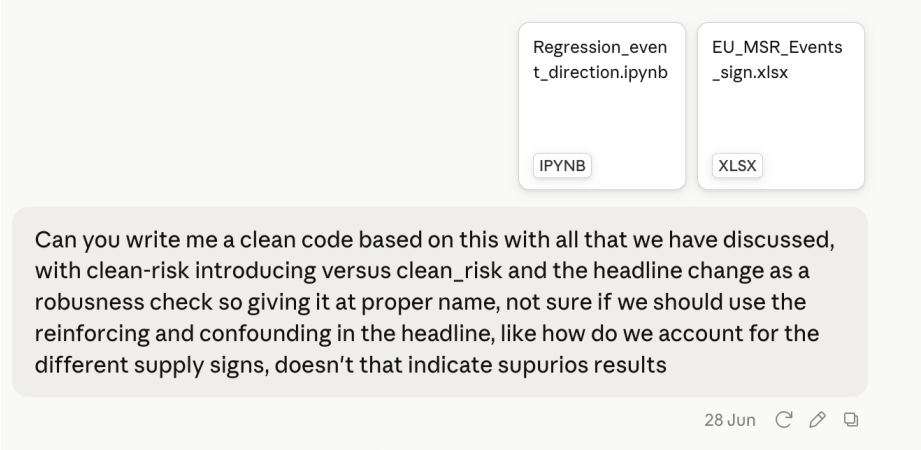
I rejected that parallel as drawn. Ramelli et al. (2021) estimate each election as a separate cross-

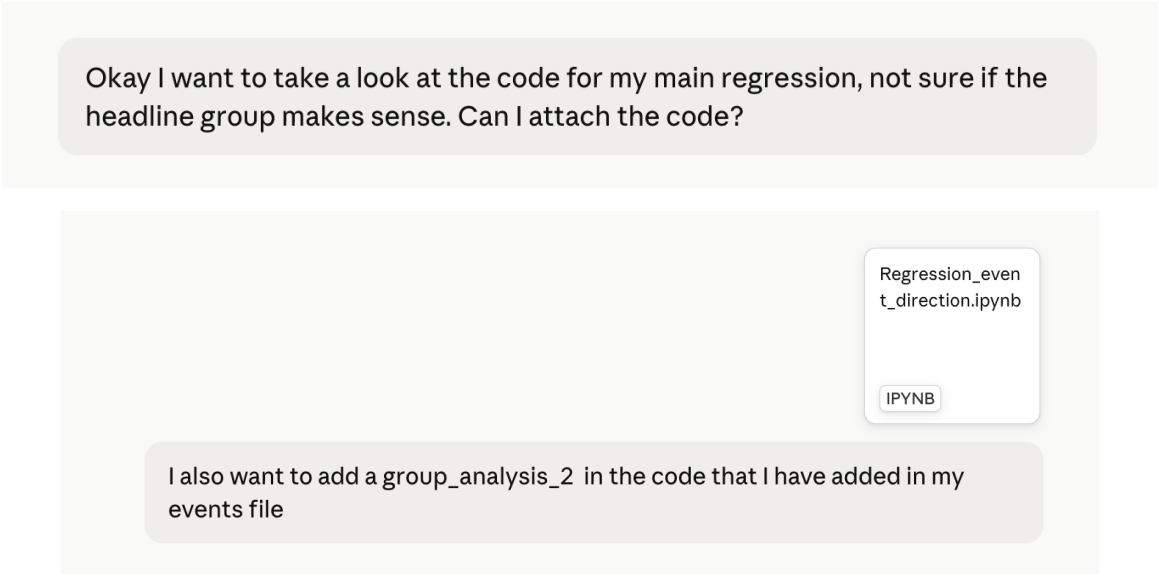
sectional event study and read the opposite-signed effect off the comparison between the two; the "sign" in their design is an interpretive contrast across two events, not a regressor within a single specification. This thesis instead encodes direction as an explicit pooled sign variable (`pv_ar_sign`) across many legislative events, which is exactly the step that constitutes its contribution over the two-event template. Accepting the model's equivalence would have collapsed the distinction the literature review exists to draw. I therefore kept the model's account of the opposite-shock logic and its typographical fixes, but rewrote the passage so that the opposite-shock pair is credited as what *permits* signing in a two-event setting, while the pooled sign variable is presented as this thesis's generalisation of that idea. I also acted on the verification flag, confirming that the analyst-forecast result belongs to the Wagner–Zeckhauser–Ziegler companion paper and correcting the citation accordingly rather than attaching the claim to the wrong source.

Example 2: Why the pooled specification could not be the headline

The shape of the entire results section, of clean-risk events as the identifying specification and the pooled all-signed regression demoted to a robustness figure, was settled in the exchange below, which began from a suspicion I brought to the model about the pooled "headline" regression and a request to implement a restructured grouping in code.

Example 2: Why the pooled specification could not be the headline





Key prompt and follow-ups (28 June): questioning the pooled headline group, requesting a restructured grouping in code, and proposing a new grouping column.

The model first inspected the attached workbook and pushed back on the premise of the third prompt: the refined buckets I wanted did not need a new hand-coded `group_analysis_2` column, because each was already derivable from the `analysis_group` and `event_type` fields the sheet contained, so adding one would only create a field that could drift out of sync. It then confirmed the spurious-results concern and supplied the mechanism: because the treatment `pv_ar_sign` × `carbon_z` encodes only the risk prediction while the realized abnormal return also carries the supply (cash-flow) channel which itself loads on carbon intensity. Therefore, supply is an omitted variable correlated with the regressor on event dates. Consequently, it biases the pooled coefficient away from zero for events whose supply and risk signs agree (reinforcing) and toward zero for those where they oppose (confounded); in the fourteen-event pool these opposing biases are present at once, so the pooled coefficient is an uninterpretable average rather than a risk-channel estimate. The model concluded that the pooled specification could not be repaired by better pooling: it had to be demoted to an explicitly conflated robustness figure, with the supply-neutral clean-risk set (split into sign-unambiguous introducing and sign-sensitive resolving events) promoted to the

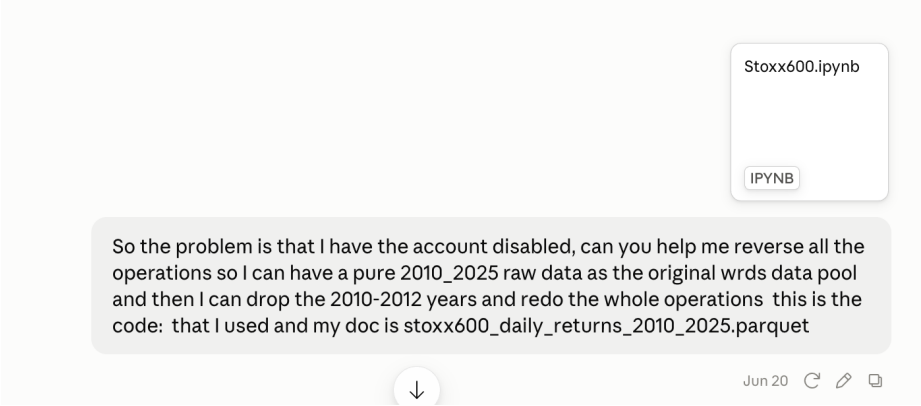
primary specification and produced clean code implementing that hierarchy.

I adopted the restructured hierarchy because it confirmed and made precise the worry that had prompted the exchange: a pooled regression mixing events with different supply signs cannot be read as a risk-channel estimate. This is the decision that fixed the identification strategy for the whole thesis. I also dropped my own plan to add a `group_analysis_2` column once the model showed the distinctions were already encoded in the workbook, since the extra field would only have created a synchronisation burden for no gain. What I did not take on trust was the generated code: I checked it against the membership and sign diagnostic the model printed, confirming that each event fell into the intended bucket and that every clean-risk event carried `supply_ar_sign` $= 0$ before using the output. The substantive argument here, the suspicion that the pooled specification was uninterpretable, was mine; the exchange's value was in stress-testing it until it became a well-drawn argument instead of an intuition-based one.

Example 3: Rebuilding the abnormal-return construction under a data-access constraint

Midway through the project my WRDS account was disabled, which meant I could no longer re-pull data. Restoring my account access would have taken a few days and therefore would have caused me to lose time. The rolling beta's were previously constructed without using the 2010-2012 data which I had realised it was the wrong approach. As such the AR (the dependent variable for every regression in this thesis) had to be re-built from a multi-step pipeline: merge the momentum factor, convert the full 2010–2025 return panel to USD excess returns, estimate six-factor rolling betas lagged one day, difference realised from predicted returns, and trim the warm-up years out only afterwards. I asked the model to help me reverse my earlier operations back to a clean raw sample and rebuild from there, attaching my code and naming the file I had saved.

Example 3: Rebuilding the abnormal-return construction under a data-access constraint



Key prompt: under the disabled-account constraint, asking the model to recover a clean raw 2010–2025 sample and rebuild the construction, with the code attached.

The model did not simply do as asked. It first read the attached notebook to see what my code had actually done to the data, then reframed the problem: I did not need to reverse any operations or re-pull from WRDS at all. The one raw file I already held was sufficient; the momentum factor is a free public download from the Dartmouth data library rather than a WRDS product, so the disabled account did not block it; and the currency conversion only had to be re-run on the full 2010–2025 frame instead of the 2013-onward cut I had saved. It delivered a single parameterised script for the whole construction and, before handing it over, unit-tested the rolling-beta and one-day-lag mechanics on synthetic data with known loadings confirming that the routine recovered the true betas and never used a day's own return in its own prediction.

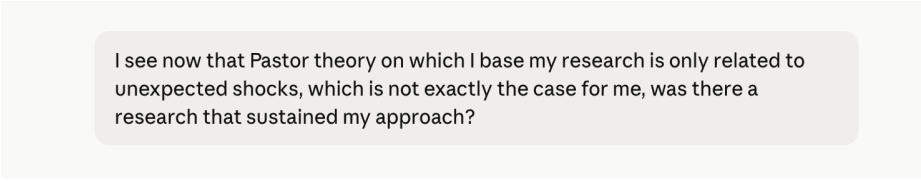
Because this was delegated code producing the thesis's dependent variable under a constraint I could not easily re-check against a fresh data pull, I verified the output independently before relying on it. I accepted the reframing: no re-pull, keep the warm-up years, re-run the conversion on the full frame. Each step was something I could confirm against files I already held rather than take on trust. I then ran the validation the design required: the mean abnormal return was approximately zero and roughly symmetric with the mean -1.25 bp and median -1.12 bp coverage confirmed the series

effectively begins in November 2013, so the January 2014 event falls after the warm-up boundary (the reason for the 48-month rather than 60-month window); and a spot-check of a known firm (BP) returned an economically sensible market beta of 0.87, with abnormal returns that spiked on its own earnings dates rather than being flat. I also confirmed that no abnormal-return rows survived in the 2010–2012 warm-up years. Only then did I merge the series into the event panel.

Example 4: Testing the theoretical framework against my own design

The thesis rests on the Pástor–Veronesi framework, which is usually described in terms of *unexpected* shocks. Because MSR decisions sit on a public legislative calendar and are anticipated well in advance, I was unsure whether the framework applied to my setting at all and asked the model whether any research supported the extension.

Example 4: Testing the theoretical framework against my own design



Key prompt: questioning whether the Pástor–Veronesi framework applies to scheduled, anticipated MSR events, and asking for supporting literature

The model first corrected the premise. Pástor and Veronesi (2012) does not require the event to be a surprise in its timing; it requires residual uncertainty about the *outcome* to be resolved at the announcement, and that date can be known far in advance. It then identified the literature that establishes this. Kelly et al. (2016) apply the framework to elections and summits, events with calendar dates fixed months ahead, and find the predicted pricing still holds, because what is priced is the resolution of outcome uncertainty rather than the timing; Savor and Wilson (2013) make the same point for scheduled macroeconomic releases. As the closest empirical precedent it pointed to

Hengge et al. (2023), essentially my regression, firm-level abnormal returns interacted with carbon intensity around EU ETS events on European firms, who confront anticipation directly by reading it off a pre-event coefficient, exactly the role my $[-5, -2]$ window plays. Finally it sharpened rather than dissolved the concern: because a signed event dummy fires at full weight regardless of how much was anticipated, the design bounds the announcement-window carbon differential of the *residual unanticipated* component of MSR news, not the full ex-ante risk premium, and partial anticipation attenuates estimates toward zero rather than inflating them.

I adopted this, and it became load-bearing rather than a footnote. Kelly et al. (2016) now anchors the paragraph of the literature review that pre-empts the most predictable objection to the design: that MSR outcomes were calendar-scheduled and priced in before the window and the residual-unanticipated-component scoping runs through how the thesis states what its null does and does not bound. I did not, however, take the model's characterisation of the source on trust: its one-line summary of what Kelly et al. (2016) show was its own compression, so I read the paper to confirm the scheduled-event result is stated as claimed before citing it, and I kept the attribution precise: the framework validates the scheduled-event application, whereas the cross-sectional carbon interaction I estimate is the applied literature's operationalisation of it, not a formal proposition of the original model. What the exchange changed was my understanding of what my own design identifies.

Reflection

Using AI while writing my master thesis has been transformative. Like each technological revolution before it, and I do believe this to be a technological revolution, it has consequences that are easy to predict and others that are perhaps still to come. The gain in efficiency is indisputable. GenAI can check which source I am thinking of when referencing an idea from a paper in seconds, work that could have taken me hours. When my WRDS access was withdrawn and the abnormal return construction had to be rebuilt around that constraint, the redesign was compressed into a fraction of the time it would otherwise have taken (Example 3). The price for this speed is difficult to measure.

Were I to search manually for the idea I wanted to reference, might I have come across another idea, previously forgotten, made a new connection and enriched my thesis through it? It is possible. Evans (2008) found that when journals moved online and search replaced browsing, citation patterns narrowed instead of widening. This is a question of opportunity cost, and I think there is a strong area of research that should develop on how and when researchers should use AI.

Furthermore, when discussing explicitly GenAI and writing, the question of style is at the forefront. GenAI, when prompted well, is capable of very precise writing. From the Greeks we know that persuasion has ethos, logos and pathos (Aristotle, Rhetoric I.2). I would separate the three, because each is limited differently. Logos is where models are currently weakest, not strongest. Dahl, Magesh, Suzgun, and Ho (2024) find hallucination rates of at least 58% on law-specific questions, and the models are poor at predicting their own errors. But this is a limitation of grounding, and I expect it to be solved. Ethos is a different case. Aristotle places it in the speech, and a model can produce the markers of credibility. What it cannot produce is an author who can be wrong and be discredited for it. In this thesis the ethos is mine, because I am the one examined on the claims. Pathos is the open question, and it is what distinguishes excellent writing from good writing. While Finance is mostly a technical discipline, there are examples of economic writing that moves. Guardrails for AI-assisted writing have to be established in order to protect the most valuable writing at the top of the pyramid.

There is also a limit that is harder to see. A language model estimates a distribution over text and generates by sampling from it, and nothing in it reaches beyond the statistical structure of what it was trained on. It is tempting to conclude that novel thought is ruled out, but this is too quick, since any estimator that generalises produces outputs it never observed. The difficulty is that the training corpus is not observable. When the output looks original, it is difficult to differentiate extrapolation from retrieval. It is an identification problem and the counterfactual is missing. I saw a smaller version of this in my own work, when the headline event grouping was demoted because supply-sign contamination made it unusable as a test of the risk channel (Example 2). The argument is easy to produce once the contamination is named. Naming it required knowing what the taxonomy was for.

From an ethical point of view, the use of AI has immediate consequences for resource use. For example, trading long-term resource management for convenience: the International Energy Agency (2025) puts data centre electricity consumption at around 415 TWh in 2024, roughly 1.54% of the global total, and expects it to more than double by 2030. I am of the belief that technological progress is irreversible. Hence, I will conclude by saying that we are adjusting to a huge jump in capability, and the way forward is to move consciously, discerning when AI improves the work and where it replaces the part of the work that was the point.

B Variable Definitions and Event Set

B.1 Variable Definitions

This appendix defines every variable entering the estimation, together with its source and its construction. Three naming conventions appear in the thesis and are reconciled here: the symbol used in the equations of Section 6, the shorthand used in the table notes, and the field name in the estimation panel (`stoxx600_panel_step5.parquet`). Where the constructions differ in more than notation (the raw, logged and standardised carbon measures, and the two event signs), each is listed separately. The computation of the effect bound r is documented separately in Appendix C.1.

Table 6: Variable definitions, construction, and sources

Symbol	Panel field	Definition and construction	Source
Panel A. Dependent variable			
$AR_{i,t}$	`ar`	Daily abnormal return: the residual from the six-factor model of equation (7), formed as realised excess return minus its factor-model expectation, equation (8). Reported throughout in basis points. Loadings are estimated on a trailing 48-month (1,008 trading-day) rolling window ending at $t-1$ and lagged one further day within firm, so the benchmark uses no information dated t or later.	Constructed; `statsmodels` `RollingOLS`
$R_{i,t} - R_{f,t}$	`excess_return_usd`	Daily total return in USD less the daily risk-free rate. Total returns are built from closing price, cumulative adjustment factor and total-return factor (`prccd`, `ajexdi`, `trfd`) and converted to USD at the daily spot rate; firm-days with a missing conversion rate are dropped.	Compustat Global Security Daily (WRDS); validated against FactSet
Panel B. Factor model inputs			
$R_{m,t} - R_{f,t}$, SMB_t, HML_t, RMW_t, CMA_t	—	Daily European Fama–French five-factor returns: market excess return, size (SMB), value (HML), profitability (RMW) and investment (CMA).	Kenneth R. French Data Library (European daily factors)
MOM_t	—	Daily European momentum (winners-minus-losers) factor, merged from the same library and appended to the five-factor set to give the six-factor model.	Kenneth R. French Data Library

Table 6, continued

Symbol	Panel field	Definition and construction	Source
$R_{f,t}$	—	Daily risk-free rate accompanying the European factor set.	Kenneth R. French Data Library
$\hat{\beta}_{ki,t-1}$	—	Firm-specific factor loadings, $k = 1, \ldots, 6$, estimated on the trailing window ending at $t-1$. Estimation runs on 2010–2025 so that betas are warm by the first event (January 2014); the resulting AR series is trimmed to 2013 onward, so the 2010–2012 years serve only as warm-up and never enter the results.	Constructed
Panel C. Treatment variables			
s_t	`pv_ar_sign`	Signed regulatory-risk indicator, written s^{pv} in Table 7, $\in \{-1, 0, +1\}$. Equal to -1 on risk-introducing event windows, $+1$ on risk-resolving windows, and 0 otherwise. Introducing events take -1 because required returns rise while the realised announcement return is negative: the price fall is what installs the higher premium.	Hand-coded from the event taxonomy (Table 7)
s_t^{sup}	`supply_ar_sign`	Signed cash-flow indicator, $\in \{-1, 0, +1\}$, signing supply-active events by the direction in which they move expected compliance costs rather than regulatory risk. Used in the supply-placebo specification to give the competing channel its own treatment.	Hand-coded from the event taxonomy (Table 7)

Table 6, continued

Symbol	Panel field	Definition and construction	Source
$s_t \cdot z_{i,t}$	`snx`	The interaction term and the coefficient of interest in equation (9). Its coefficient β is read in basis points of daily abnormal return per one within-year standard deviation of log carbon intensity. A positive β places high-carbon firms in the amplification direction.	Constructed
—	—	*Event windows.* The announcement window is the three trading days $[-1, +1]$ around the event date; the pre-event window is $[-5, -2]$. The two are disjoint, and the 27 event windows are non-overlapping, so no firm-day carries two signs.	Constructed
Panel D. Carbon intensity			
—	—	*Raw carbon intensity.* Scope-1 emissions in tonnes of CO_2 divided by revenue, at the firm-year level. Firm-years reporting zero or negative intensity are dropped, since the ratio cannot be logged. Emissions are the Refinitiv field `TR.CO2DirectScope1`.	Refinitiv/LSEG
—	—	*Log carbon intensity.* Natural logarithm of the raw ratio, taken because the raw measure spans nine orders of magnitude and would otherwise be identified off a handful of extreme emitters (Figure 1).	Constructed

Table 6, continued

Symbol	Panel field	Definition and construction	Source
$z_{i,t}$	`carbon_intensity_z`	Within-year cross-sectional z-score of log carbon intensity, written carbon$_{z,it}$ in the table notes. For each calendar year, log intensity is demeaned and scaled by that year's cross-sectional standard deviation, so a firm is ranked against contemporaneous peers rather than against the pooled sample (Figure 2). Merged to returns with a one-year lag (`merge_asof`), so the classification on any event date uses only information already public.	Constructed
—	—	*Carbon quartile.* Top-minus-bottom carbon-quartile indicator replacing $z_{i,t}$ in one robustness row. Its coefficient is scaled per quartile contrast, not per standard deviation, and is not comparable in magnitude to the other rows.	Constructed
—	—	*Reported-emissions flag.* Indicator that a firm-year's Scope-1 value is company-disclosed rather than vendor-modelled, used to restrict the sample in the data-provenance robustness check following Aswani et al. (2024). The *Median*, *CO2* and *Energy* estimation models are treated as vendor-estimated. The flag is the Refinitiv field `CO2EstimationMethod`.	Refinitiv/LSEG

Table 6, continued

Symbol	Panel field	Definition and construction	Source
Panel E. Controls and fixed effects			
—	`energy_ret`	Daily return on Brent crude, the European crude benchmark. The price series is forward-filled onto the panel's European trading-day grid before differencing, so that an event date falling in a gap in the source calendar is retained rather than dropped; `energy_ret` is approximately zero on such a day. The EU allowance price is deliberately *not* used here: it lies on the causal path of the MSR events under study, so conditioning on it would partly absorb the treatment.	FRED, series `DCOILBRENTEU` (Crude Oil Prices: Brent—Europe)
—	`energy_x_carbon`	`energy_ret` × `carbon_intensity_z`. Added in one robustness row to absorb the single confound the day fixed effect cannot: a time-varying energy-price move whose loading differs across the carbon cross-section on event dates.	Constructed
α_i	`isin`	Firm fixed effect, absorbing time-invariant firm characteristics including the average level of carbon intensity.	Constructed
δ_t	`date`	Day fixed effect, absorbing every market-wide shock on an event date ,including the factor realisations themselves, so that β is identified purely from the cross-sectional gradient in AR on the same day.	Constructed

Table 6, continued

Symbol	Panel field	Definition and construction	Source
γ	—	Coefficient on the level of $z_{i,t}$. It does not vanish under firm fixed effects because within-year re-standardisation induces year-to-year drift in a firm's z-score even when its underlying intensity is unchanged.	Constructed

Notes. Estimation sample: historical STOXX 600 constituents, 2013–2025. The index membership panel is unbalanced by construction so as to avoid survivorship bias: 1,128 firms appear over 2010–2025, of which 1,052 accumulate at least one abnormal-return observation after the rolling-beta warm-up, and 850 also carry Refinitiv carbon data and constitute the final estimation sample (8,350 matched firm-years; 1.92–1.95 million firm-days across specifications). Twenty-seven MSR events are hand-classified, of which fourteen carry a non-zero sign; the full listing is Table 7. Estimation uses `pyfixest` with two-way clustering on `isin` and `date` for the pooled specifications and firm clustering for the event subsets, for the reasons given in Section 6.5.

B.2 MSR Events

Table 7: MSR regulatory events: full classification

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
1	2014-01-22	**Commission proposes the MSR.** A new ETS instrument is tabled with uncertain final parameters and start date. Regulatory uncertainty about whether, when and how the MSR will operate enters the market.	Reinforcing	-1	-1

Continued on next page

Table 7 — continued from previous page

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
2	2015-03-05	**Council General Approach on MSR.** Council settles its negotiating position. Partial resolution: political feasibility and approximate start date become clearer, though Parliament still pushed for earlier operation.	Clean-risk	$+1$	0
3	2015-10-06	**Decision (EU) 2015/1814 adopted.** Major resolution event. MSR formally established with 12% intake rate, 100m minimum annual placement, operational start 1 Jan 2019. Core parameters fixed in law.	Confounded	$+1$	-1
4	2016-04-04	**Poland files Case C-5/16 at the ECJ.** Legal/annulment risk enters: Poland challenges the validity of the MSR Decision. Until ruling, residual probability that the instrument could be struck down or narrowed.	Clean-risk	-1	0
5	2017-05-15†	**TNAC 2016 publication (first TNAC).** First annual TNAC publication; MSR not yet operational. Day-of-month not independently verified; placeholder set to 15 May to match the then-applicable statutory deadline.	Neutral	0	0
6	2018-03-14	**Directive (EU) 2018/410 adopted.** Resolution and regime strengthening. Intake rate doubled to 24%, minimum placement to 200m, invalidation rule introduced from 2023. Direction of the surprise was stricter, useful for asymmetric-response modelling.	Confounded	$+1$	-1
7	2018-05-15†	**TNAC 2017 publication.** Annual TNAC publication for 2017; MSR not yet operational. Day-of-month not independently verified; placeholder set to 15 May.	Neutral	0	0

Continued on next page

Table 7 — continued from previous page

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
8	2018-06-21	**ECJ judgment in Case C-5/16.** Legal risk resolved in favour of the Commission. Court confirms MSR is a lawful, one-off intervention to correct a structural ETS weakness.	Clean-risk	$+1$	0
9	2019-01-01	**MSR becomes operational.** End of pure anticipation phase. Implementation risk replaced by operational risk: rules now bite, intake calculations begin.	Neutral	0	0
10	2019-05-15†	**TNAC 2018 publication.** Annual TNAC publication for 2018; first under the operational MSR. Day-of-month not independently verified; placeholder set to 15 May.	Neutral	0	0
11	2019-12-11	**European Green Deal Communication.** Broad regulatory risk shock for the EU ETS ecosystem. Signals further tightening (later Fit for 55), re-opening MSR parameters for revision.	Reinforcing	-1	-1
12	2020-05-15†	**TNAC 2019 publication.** Annual TNAC publication for 2019. Surplus $\sim$1.39 bn allowances. Day-of-month not independently verified; placeholder set to 15 May.	Neutral	0	0
13	2021-05-15†	**TNAC 2020 publication.** Annual TNAC publication for 2020 (COVID year). Day-of-month not independently verified; placeholder set to 15 May.	Neutral	0	0
14	2021-07-14	**Fit for 55 package – MSR proposal.** Major regulatory risk: MSR parameters re-opened (extension of 24% rate, refined TNAC formula, invalidation threshold) and ETS2 proposed as a parallel system with its own MSR rules.	Reinforcing	-1	-1

Continued on next page

Table 7 — continued from previous page

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
15	2022-05-13	**TNAC 2021 publication.** TNAC = 1,449m allowances. 347.8m placed in MSR over 1 Sep 2022 – 31 Aug 2023.	Neutral	0	0
16	2022-06-22	**EP plenary vote on ETS revision fails.** Elevated regulatory risk: first plenary vote rejected, file sent back to committee. Political deal in doubt for several weeks. A rare process-failure event useful for identification.	Clean-risk	-1	0
17	2022-06-29	**Council General Approach on ETS revision.** Partial resolution: Council position fixes a baseline for trilogues. Reduces uncertainty over the floor of ambition.	Clean-risk	$+1$	0
18	2022-12-18	**Trilogue provisional agreement on ETS revision.** Substantive resolution: text of the ETS/MSR revision agreed in principle. Most of the regulatory uncertainty over the Fit for 55 MSR rebuild collapses here, ahead of formal adoption.	Confounded	$+1$	-1
19	2023-01-01	**First MSR invalidation ($\sim$2.5 bn allowances).** Mechanically determined under existing rules, but salient regulatory event for high-exposure firms — the largest single removal of allowances in EU ETS history.	Supply placebo	0	-1
20	2023-04-19	**Decision (EU) 2023/852 adopted.** Resolves uncertainty on intake parameters: 24% rate and 200m minimum extended through 31 Dec 2030; reversion to 12% / 100m only from 1 Jan 2031.	Confounded	$+1$	-1

Continued on next page

Table 7 — continued from previous page

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
21	2023-05-10	**Directive (EU) 2023/959 adopted.** Final resolution of Fit for 55 MSR rebuild: invalidation threshold fixed at 400m (from 2024), threshold-effect mitigation mechanism, aviation in TNAC, TNAC publication deadline moved from 15 May to 1 June.	Risk-ambiguous	$+1$	0
22	2023-05-15	**TNAC 2022 publication.** TNAC = 1,134.8m allowances. 272.4m placed in MSR over 1 Sep 2023 – 31 Aug 2024. First invalidation year.	Neutral	0	0
23	2024-01-01	**Second invalidation; revised rules take effect.** 381m allowances invalidated. From this date the 400m fixed threshold, threshold-effect mitigation, and aviation inclusion apply — a regime-change moment within the operational phase.	Supply placebo	0	-1
24	2024-05-31	**TNAC 2023 publication.** First publication under the new 1 June deadline. Aviation now included in TNAC calculation.	Neutral	0	0
25	2025-01-01	**Third annual invalidation.** 270.5m allowances invalidated under the now-standard rule. Repeated invalidations are useful repeated events in panel analysis.	Supply placebo	0	-1
26	2025-05-28	**TNAC 2024 publication.** TNAC = 1,148m allowances. 275.5m placed in MSR over 1 Sep 2025 – 31 Aug 2026.	Neutral	0	0

Continued on next page

Table 7 — continued from previous page

ID	Date	Event and description	Analysis group	s^{pv}	s^{sup}
27	2025-11-27	**Commission proposes ETS2 MSR amendments.** New regulatory risk introduced for ETS2: price-triggered top-up (double injection if EUA price >EUR 45/tonne in 2020 prices, up to 80m/year until end-2029), validity of ETS2 allowances extended beyond 2030, additional buffer for earlier intervention.	Confounded	-1	$+1$

Notes. Complete listing of the 27 MSR-related events hand-classified by the author from the legislative record (Commission proposals, Council and Parliament positions, adopted Decisions and Directives, ECJ filings and judgments, and annual TNAC publications) and cross-validated against the news-intensity series. Descriptions are the author's own classification notes. s^{pv} is the regulatory-risk sign, coded -1 for risk-introducing events and $+1$ for risk-resolving events; s^{sup} is the supply (cash-flow) sign, coded -1 where the event tightens expected allowance supply and $+1$ where it loosens it. Clean-risk events are supply-neutral ($s^{sup} = 0$) and form the primary identifying set; Risk-ambiguous events are also supply-neutral but do not carry a clean risk direction, and are held out of the clean-risk set for that reason: confounded events are ambiguous as to which channel moves returns, risk-ambiguous events as to direction within the risk channel; confounded events carry non-zero signs on both channels; reinforcing events are those on which the two signs coincide; supply-placebo events are risk-neutral ($s^{pv} = 0$) but supply-active; neutral events receive no event window and remain in the control pool.

† Dated to the month only: the publication day could not be independently verified, and the date shown is the then-applicable statutory deadline. These five events are excluded from the $[-1, +1]$ and $[-5, -2]$ event windows, which require exact dating. All five are risk- and supply-neutral, so their exclusion leaves every treated event set unchanged.

C Statistical Appendix

C.1 Computation of the effect bound

Each null in the results is summarised not by a p-value but by the magnitude of carbon amplification the data still permit. The logic is test inversion. For a specification with estimated interaction

Table 8: Company-reported emissions only: carbon-scaled abnormal-return response versus the full sample

			Announcement $[-1,+1]$		Pre-event $[-5,-2]$	
Specification	Cluster	Events	Full	Reported	Full	Reported
Clean-risk (all)	2-way	5	0.80 (4.37)	0.95 (4.57)	-5.38^{*} (3.16)	-5.44 (3.28)
Introducing	firm	2	4.61 (3.22)	5.54 (3.27)	-0.10 (2.66)	0.25 (2.70)
Resolving	firm	3	-1.83 (2.80)	-2.20 (2.85)	$-9.02^{\dagger}$ (2.04)	$-9.35^{\dagger}$ (2.06)
Supply placebo	firm	3	3.42 (3.37)	3.54 (3.37)	1.71 (1.09)	1.52 (1.09)
Reinforcing	firm	3	-0.60 (2.82)	-0.93 (2.92)	0.47 (1.83)	0.45 (1.83)
Pool all signed	2-way	14	0.58 (2.45)	0.89 (2.47)	-0.45 (2.11)	-0.98 (2.03)

Notes. Each cell reports the coefficient β on sign $\times$ carbon$_z$ from the specification in Table 4, in basis points of daily abnormal return per within-year standard deviation of log Scope-1 carbon intensity; standard errors in parentheses. *Full* reproduces the main-sample columns; *Reported* restricts the estimation sample to firm-years whose Scope-1 value is company-disclosed (Refinitiv `CO2EstimationMethod` = *Reported*), dropping the vendor-estimated *Median*, *CO2*, and *Energy* models. The restriction removes ≈2.5% of carbon-scaled firm-day observations; the event counts are unchanged.

Stars mark two-way (firm + date) clustered rows only: $^{*}\,p<0.10$, $^{**}\,p<0.05$, $^{***}\,p<0.01$. The pooled clean-risk pre-event coefficient is marginal in both samples ($p=0.09$ full, $p=0.10$ reported); the reported column falls just short of the 10% threshold but is unchanged in substance. Firm-clustered rows are read as point estimates against their confidence intervals, not as significance tests.

coefficient $\hat{\beta}$ and clustered standard error $\mathrm{se}(\hat{\beta})$, the two-sided 5% test of the hypothesis $\beta=\beta_0$ rejects when $|\hat{\beta}-\beta_0|/\mathrm{se}(\hat{\beta})>z_{0.975}=1.96$. The 95% confidence interval is exactly the set of hypothesised values that survive this family of tests,

$$\left\{\,\beta_0\;:\;\frac{|\hat{\beta}-\beta_0|}{\mathrm{se}(\hat{\beta})}\le 1.96\,\right\}\;=\;\big[\,\hat{\beta}-1.96\,\mathrm{se}(\hat{\beta}),\;\;\hat{\beta}+1.96\,\mathrm{se}(\hat{\beta})\,\big], \tag{10}$$

so every value outside the interval is rejected by the observed data at the 5% level. Because $\hat{\beta}\neq 0$, the interval is asymmetric about zero; a claim about the admissible *magnitude* of the effect must therefore use the interval endpoint farthest from zero,

$$r\;=\;|\hat{\beta}|\;+\;1.96\,\mathrm{se}(\hat{\beta}), \tag{11}$$

the specification's *effect bound*: every hypothesised response with $|\beta_0|>r$ is rejected by the observed data. The normal critical value is used throughout; for the two-way-clustered specifications

the number of date clusters is large enough that the difference from a cluster-t critical value is immaterial.

The bound is reported specification by specification, computed only on rows whose standard errors are reliable: the two-way (firm + date) clustered rows and only for the announcement window. The headline figure is the primary clean-risk specification's, $r = 0.80 + 1.96 \times 4.37 = 9.4$ bp. The remaining two-way announcement-window rows give $r = 5.4$ bp (pooled all-signed), 10.3 bp (60-month beta window), 9.1 bp (energy-return $\times$ carbon control), 9.9 bp (reported-only clean-risk) and 5.7 bp (reported-only pooled all-signed), so no reliable specification widens the headline bound by more than one basis point. The quartile row is excluded because its coefficient is scaled per top-minus-bottom carbon quartile rather than per standard deviation and is not measured on the same axis. Where a single round figure is convenient in the text, 9.4 bp is rounded *outward* to ± 10 bp: rounding a bound toward zero would assert the exclusion of effects the interval does not exclude, so rounding must always widen the bound, never tighten it.

The firm-clustered subsets are excluded from this maximum by construction. Firm-only clustering sets to zero the within-date cross-sectional correlation of high-carbon firms' abnormal returns; because that correlation is positive, the resulting standard error is a lower bound and the r it implies understates the interval's true reach. Bounding the design's precision off an understated standard error would overstate that precision. These subsets are instead read as point estimates; their announcement-window coefficients of 4.61, -1.83, 3.42, and -0.60 bp all lie below 5 bp in magnitude and corroborate the near-zero conclusion without entering the bound.

Two further statistics delimit what the bound does and does not say. First, the bound is a *post-data* statement conditional on the realised $\hat{\beta}$ and $\mathrm{se}(\hat{\beta})$, which values are rejected, and is distinct from the *ex-ante* question of what the design could have detected. The latter is answered by the minimum detectable effect at conventional 80% power,

$$\mathrm{MDE} \;=\; \big(z_{0.975} + z_{0.80}\big)\,\mathrm{se}(\hat{\beta}) \;=\; (1.96 + 0.84) \times 4.37 \;\approx\; 12.2 \text{ bp}, \tag{12}$$

the smallest true effect the pooled clean-risk specification would have rejected zero against with probability at least 0.80. A true carbon amplification of that size or larger would, with high probability, not have produced the observed null. Second, the bound is formally an equivalence statement: declaring the effect to lie within $\pm\Delta$ by two one-sided tests at level α is equivalent to the $(1-2\alpha)$ confidence interval lying inside $[-\Delta, \Delta]$. The convention here, requiring the 95% interval, rather than the 90% interval of the standard two-one-sided-tests procedure, to sit inside the bound runs each one-sided test at 2.5% and is the stricter variant.

The minimum detectable effect is an ex-ante property of the design rather than a confidence bound, and is not used as one anywhere in the text. For the same reason every bound reported here is scoped to the announcement window: the pre-event intervals are wider, reaching 12.3 bp on the two-way-clustered rows, and are read as an anticipation diagnostic rather than as a bound on anything.

C.2 Attenuation: from the structural reaction to the estimated coefficient

The estimated coefficient stands at the end of a chain of objects,

$$\beta_{\text{struct}} \xrightarrow{\lambda_{\text{pers}},\,\lambda_{\text{ant}}} \beta_{\text{win}} \xrightarrow{\lambda_{\text{ME}},\,\lambda_{\text{sign}}} \operatorname{plim} \hat{\beta}, \tag{13}$$

where β_{struct} is the structural reaction (defined below), β_{win} the true announcement-window estimand, and the four factors $\lambda_k \in (0, 1]$ attenuate in one direction only. The first pair is economic: it shrinks the true window response itself. The second pair is econometric: it shrinks the estimate relative to that response. This subsection derives each factor and their composition. Throughout, calculations condition on event-window firm-days, on which the signed indicator satisfies $s_t^2 = 1$; off-window days identify the fixed effects and the carbon level term, not β.

Regime persistence. For a claim paying a single terminal payoff $D_{i,T}$, the price at τ satisfies the identity $P_{i,\tau} = \mathbb{E}_\tau[D_{i,T}]\, e^{-(r+\pi_i)(T-\tau)}$, which defines the premium π_i as the excess part of the

continuously compounded expected return. Taking logs and differencing across an announcement that leaves expected payoffs unchanged, the defining property of a supply-neutral event, gives

$$\Delta \ln P_{i,\tau} \;=\; -(T-\tau)\,\Delta\pi_i. \tag{14}$$

Let the premium revision load on carbon exposure, $\Delta\pi_i = \phi\, z_i\, \Delta\bar{\pi}$, with $\Delta\bar{\pi}$ the market-wide revision the event delivers; the carbon-scaled reaction is then $AR_i = -(T-\tau)\,\phi\,\Delta\bar{\pi}\, z_i$. Defining β_{struct} as the reaction to a fully surprising shock of typical sample-event magnitude at a reference horizon H (both normalisations discussed below),

$$\beta_{\text{win}} \;=\; \frac{T-\tau}{H}\,\beta_{\text{struct}} \;\equiv\; \lambda_{\text{pers}}\,\beta_{\text{struct}}, \qquad \lambda_{\text{pers}} \in (0,1]. \tag{15}$$

An event arriving with little of the regime's life remaining can reprice only the premium accruing over that remainder, however large the per-period premium.[3]

Partial anticipation. Prices respond to news; the event indicator responds to dates. Let $a \in (0,1]$ be the share of an event's information content arriving inside the announcement window, the complement having diffused earlier given that the Market Stability Reserve is a forecastable rule, its legislative milestones scheduled and its mechanics deterministic. The return process inside the window is driven by the news innovation $a\,s_t$, so $AR_{it} = \beta_{\text{win}}\,a\,(s_t z_i) + e_{it}$, while the regression projects on $s_t z_i$ at full weight:

$$\operatorname{plim}\hat{\beta} \;=\; \frac{\operatorname{Cov}\big(AR,\; sz\big)}{\operatorname{Var}(sz)} \;=\; \frac{\beta_{\text{win}}\,a\,\operatorname{Var}(sz)}{\operatorname{Var}(sz)} \;=\; a\,\beta_{\text{win}} \;\equiv\; \lambda_{\text{ant}}\,\beta_{\text{win}}. \tag{16}$$

The coefficient recovers news per indicator, not news per event. If diffusion is smooth over months rather than concentrated immediately pre-announcement, the anticipated share does not accumulate

[3]For revisions to policy *uncertainty*,the object the clean-risk events move, the compression is stronger: in the Pástor–Veronesi log-price structure the posterior variance of the policy impact enters in proportion to the squared remaining horizon, so uncertainty revisions scale with $(T-\tau)^2$ rather than $(T-\tau)$. The linear factor in equation (15) is therefore the weaker of the two cases.

in the $[-5, -2]$ window either which is consistent with the absence of pre-event response on the sign-unambiguous introducing events.

Measurement error in carbon intensity. Let true exposure be z^* and the observed measure $z = z^* + u$ with u orthogonal to (z^*, s, ε), $\mathrm{Var}(z^*) = \sigma_*^2$, $\mathrm{Var}(u) = \sigma_u^2$. With true model $AR = \beta_{\text{win}}\, sz^* + \varepsilon$ and regressor sz, using $s^2 = 1$ on window days,

$$\mathrm{Cov}(AR,\, sz) = \beta_{\text{win}}\, \sigma_*^2, \qquad \mathrm{Var}(sz) = \sigma_*^2 + \sigma_u^2, \tag{17}$$

$$\operatorname{plim} \hat{\beta} \;=\; \beta_{\text{win}}\, \frac{\sigma_*^2}{\sigma_*^2 + \sigma_u^2} \;\equiv\; \lambda_{\text{ME}}\, \beta_{\text{win}}, \tag{18}$$

the reliability ratio. Because carbon intensity enters as a within-year cross-sectional z-score, $\mathrm{Var}(z) = \sigma_*^2 + \sigma_u^2 = 1$ by construction, so $\lambda_{\text{ME}} = 1 - \sigma_u^2$: the reliability is the signal share of the observed unit variance. Two features of the design are relevant. The firm fixed effect barely demeans the interaction, the firm-level time average of $s_t z_{it}$ is approximately zero because $s_t = 0$ on almost every day, so the relevant reliability is that of the cross-sectional variation on event dates, and the usual result that within-transformations worsen attenuation does not bite. And vendor-*estimated* emissions violate the orthogonality of u, since estimation models impute from size and sector (Aswani et al., 2024); the appropriate response is not an analytic correction but the re-estimation on firm-reported observations, which drives σ_u^2 toward zero.

Sign misclassification. Let the true sign on a treated day be s_t^* and the coded sign s_t, with misclassification probability $m = \Pr(s_t = -s_t^*)$, the flip independent of z and ε. Since $s^2 = 1$ regardless, $\mathrm{Var}(sz) = \mathrm{Var}(z)$, and

$$\operatorname{plim} \hat{\beta} \;=\; \beta_{\text{win}}\, \mathbb{E}[s_t^* s_t] \;=\; \beta_{\text{win}} \big[(1-m) - m\big] \;=\; (1 - 2m)\, \beta_{\text{win}} \;\equiv\; \lambda_{\text{sign}}\, \beta_{\text{win}}. \tag{19}$$

At $m = 0$ the signed design operates at full strength; at $m = 0.5$ the coefficient degrades to the unsigned-average cancellation the signing was built to prevent. With five clean-risk events weighted equally, one mis-signed resolving event implies $m = 0.2$ and $\lambda_{\text{sign}} = 0.6$; two imply $\lambda_{\text{sign}} = 0.2$. The introducing-only subset sets $m = 0$ by construction, since the risk-introducing -1 coding is unambiguous.

Composition and identification. Stacking the four mechanisms in one data-generating process—true exposure z^* observed with error, true sign s^* coded with flip rate m, news share a, horizon share $(T - \tau)/H$, the outcome on window days is $AR = \beta_{\text{struct}}\, \lambda_{\text{pers}}\, \lambda_{\text{ant}}\, (s^* z^*) + \varepsilon$, and projecting on $sz = s(z^* + u)$ gives, under the stated independence assumptions,

$$\boxed{\operatorname{plim} \hat{\beta} \;=\; \underbrace{(1 - 2m)}_{\lambda_{\text{sign}}}\; \underbrace{\frac{\sigma_*^2}{\sigma_*^2 + \sigma_u^2}}_{\lambda_{\text{ME}}}\; \underbrace{a}_{\lambda_{\text{ant}}}\; \underbrace{\frac{T - \tau}{H}}_{\lambda_{\text{pers}}}\; \beta_{\text{struct}} \;=\; \lambda_{\text{total}}\, \beta_{\text{struct}}.} \tag{20}$$

The factorisation is exact given the independence assumptions (measurement error orthogonal to signs; flips orthogonal to exposure); correlated mechanisms would add cross-terms, for which no economic rationale suggests itself here. Attenuation compounds multiplicatively: four individually mild factors of 0.8 already yield $0.8^4 \approx 0.41$.

Two normalisations define β_{struct} and deserve emphasis. The sign-only coding replaces each event's unobserved revision magnitude $|\Delta\bar{\pi}_e|$ with its sample-average unit, so "one unit of shock" means an event of typical sample magnitude; magnitude heterogeneity across events has mean one under this convention and contributes noise, not bias. The reference horizon H fixes the units of λ_{pers}; rescaling H rescales β_{struct} inversely, leaving the identified product unchanged. It follows that the loading ϕ, the typical revision $\overline{|\Delta\bar{\pi}|}$, and the horizon are identified only as a product: the data cannot distinguish a large carbon loading on small shocks from a small loading on large shocks. β_{struct} is, moreover, a *reaction* coefficient, the price response to a premium revision, and not the regulatory-risk premium itself, whose level the announcement design never observes.

The bound through the chain. The confidence interval of Appendix C.1 is a statement about the left-hand side of equation (20). Ruling out $|\operatorname{plim}\hat{\beta}| > 9.4$ bp is equivalent to

$$\left|\lambda_{\text{total}}\,\beta_{\text{struct}}\right| \leq 9.4 \quad \Longleftrightarrow \quad |\beta_{\text{struct}}| \;\leq\; \frac{9.4}{\lambda_{\text{total}}}, \tag{21}$$

with the intermediate objects bounded by partial unwinding: $|\beta_{\text{win}}| \leq 9.4/(\lambda_{\text{sign}}\lambda_{\text{ME}})$. Because $\lambda_{\text{total}} \leq 1$, every step from the measured object toward the structural one loosens the bound, and none of the factors is identified by the data alone. Purely for illustration: reliability 0.8, one mis-signed resolving event ($\lambda_{\text{sign}} = 0.6$), half the news anticipated ($a = 0.5$), and three-quarters of the reference horizon remaining ($\lambda_{\text{pers}} = 0.75$) give $\lambda_{\text{total}} = 0.18$ and a structural bound of $9.4/0.18 \approx 52$ bp, more than five times the measured figure. The bound is exact for the residual announcement-window repricing and dissolves geometrically along the chain, which is why the results scope their claim to the former.

The design addresses the two econometric factors internally rather than by assumption: the introducing-only subset sets $\lambda_{\text{sign}} = 1$ by construction, and the reported-emissions robustness drives λ_{ME} toward one. The two economic factors cannot be switched off and should not be: a forecastable rule with a scheduled review has small a and finite $(T-\tau)$ by design, which is precisely why residual announcement repricing is the theoretically predicted place to find little.